

3_10xPBMC_comparison_wSTAR_noPlatelet

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(pheatmap)
library(ggpubr)
library(ggbeeswarm)
getwd()
dir.create("figures_10xPBMC")
dir.create("data_10xPBMC")
dataset_id <- "10xPBMC"
sample_id <- "pbmc8k"
colorPBMC <- "#81B29A"
mode <- "pseudobulk"

# PBMCcellltypeColors <- c("B"="#91bcca",
#                           "CD14pos_Monocytes"="#3d405b",
#                           "CD8A"="#f2cc8f",
#                           "Dendritic"="#c492a7",
#                           "FCGR3Apos_Monocytes"="#e78063",
#                           "Megakaryocytes"="#c4b3ab",
#                           "Memory_CD4_T"="#81b29a",
#                           "Naive_CD4_T"="#4d7362",
#                           "NK"="#88535a")

colorAge = c("old"="#7a646a",
             "young"="#88bce7")

colorTools = c("SoloTE"="#A4DD9B", #6fc69d",
               "Stellarscope"="#4f5d93",
               "STARsolo"="#f0df93")
```

Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall ‘SeuratObject’ as the ABI for R may have changed

'SeuratObject' was built with package 'Matrix' 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall 'SeuratObject' as the ABI for 'Matrix' may have changed

Attaching package: 'SeuratObject'

The following objects are masked from 'package:base':

intersect, t

'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

Warning message in dir.create("figures_10xPBMC"):

"'figures_10xPBMC' already exists"

Warning message in dir.create("data_10xPBMC"):

"'data_10xPBMC' already exists"

```
[2]: theme_paper <- function(base_size = 17, base_family = "") {  
  theme_minimal(base_size = base_size, base_family = base_family) +  
  theme(  
    plot.title = element_text(face = "bold", size = base_size + 2, hjust = 0.  
↪5),  
    axis.title = element_text(size = base_size),  
    axis.text = element_text(size = base_size * 0.9),  
    legend.title = element_text(size = base_size),  
    legend.text = element_text(size = base_size * 0.9),  
    panel.grid.major = element_blank(),  
    panel.grid.minor = element_blank(),  
    plot.margin = margin(10, 10, 10, 10)  
  )  
}  
theme_set(theme_paper())
```

```
[3]: # celltype annotation  
  
celltypes <- read.table("/mnt/TEdata/TEbenchmarking/data/whitelists/  
↪celltype_annotation_sub_min.tsv")  
  
celltypes$V2 <- gsub(pattern="CD14pos_Monocytes", replacement =  
↪"Monocytes_CD14pos", celltypes$V2)  
celltypes$V2 <- gsub(pattern="FCGR3Apos_Monocytes", replacement =  
↪"Monocytes_FCGR3Apos", celltypes$V2)  
celltypes$V2 <- gsub(pattern="Naive_CD4_T", replacement = "T_CD4_Naive",  
↪celltypes$V2)
```

```

celltypes$V2 <- gsub(pattern="Memory_CD4_T", replacement = "T_CD4_Memory",
  ↪ celltypes$V2)
celltypes$V2 <- gsub(pattern="CD8A", replacement = "T_CD8", celltypes$V2)
celltypes$V2 <- gsub(pattern="Megakaryocytes", replacement = "Platelet",
  ↪ celltypes$V2)

PBMCelltypeColors <- c("B"="#91bcca",
  "Monocytes_CD14pos"="#3d405b",
  "T_CD8"="#f2cc8f",
  "Dendritic"="#c492a7",
  "Monocytes_FCGR3Apos"="#e78063",
  # "Platelet"="#c4b3ab",
  "T_CD4_Memory"="#81b29a",
  "T_CD4_Naive"="#4d7362",
  "NK" = "#88535a")

```

1 Prep

```

[5]: # Import Stellarscope object
objTE_stellarscope_pseudobulk <- readRDS(paste0("data_", dataset_id, "/"
  ↪ stellarscope_", dataset_id, "_", "pseudobulk", "seuratObj_noplatelet.RDS"))
objTE_stellarscope_celltype <- readRDS(paste0("data_", dataset_id, "/"
  ↪ stellarscope_", dataset_id, "_", "celltype", "seuratObj_noplatelet.RDS"))

# Import SoloTE object
objTE_soloTE <- readRDS(paste0("data_", dataset_id, "/soloTE_", dataset_id,
  ↪ "_seuratObj_noPlatelet.RDS"))

# Import STARsolo object
objTE_STARsolo <- readRDS(paste0("data_", dataset_id, "/"
  ↪ STARsolo_", dataset_id, "_seuratObj_noPlatelet.RDS"))

```

```

[6]: objTE_stellarscope_pseudobulk
objTE_stellarscope <- objTE_stellarscope_pseudobulk

objTE_stellarscope_celltype

objTE_soloTE

objTE_STARsolo

```

An object of class Seurat
 6367 features across 1056 samples within 1 assay
 Active assay: RNA (6367 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap

An object of class Seurat
 6235 features across 1056 samples within 1 assay
 Active assay: RNA (6235 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap

An object of class Seurat
 6758 features across 1056 samples within 1 assay
 Active assay: RNA (6758 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap

An object of class Seurat
 4955 features across 1056 samples within 1 assay
 Active assay: RNA (4955 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap

```
[7]: conversionTable <- read.table("annotation/annotation_hg38_conversion_withAge.
  ↪tsv") # generated with annotation_scripts/create_annotations_hg38.Rmd
head(conversionTable)
```

A data.frame: 6 × 16

	chr	start	end	strand	sw_score	locusID	family	subfam
	<chr>	<int>	<int>	<chr>	<int>	<chr>	<chr>	<chr>
1	chr1	10469	11447	-	3612	TAR1	telo	TAR1
2	chr1	11505	11675	-	484	L1MC5a_dup3	L1	L1MC
3	chr1	11678	11780	-	239	MER5B_dup1	hAT-Charlie	MER5
4	chr1	15265	15355	-	318	MIR3_dup11	MIR	MIR3
5	chr1	18907	19048	+	239	L2a_dup29	L2	L2a
6	chr1	19972	20405	+	994	L3_dup3	CR1	L3

```
[8]: # get TEs in common between Stellarscope and SoloTE

stellarscopeTEs <- Features(objTE_stellarscope)

table(stellarscopeTEs %in% conversionTable$stellarscopeID)

soloTEs <- Features(objTE_soloTE)
# remove "SoloTE" from the name of the TEs and substitute "/" and "_" with "-"
soloTEs <- gsub("\\|", "-", soloTEs)
soloTEs <- gsub("\\_", "-", soloTEs)
soloTEs <- gsub("\\?", "", soloTEs)

table(soloTEs %in% conversionTable$soloteID)

TEsInStellarscope_SoloTE <- intersect(
  ↪conversionTable$locusID[conversionTable$stellarscopeID %in% stellarscopeTEs],
  ↪conversionTable$locusID[conversionTable$soloteID %in%
  ↪soloTEs] )
```

```
length(TEsInStellarscope_SoloTE)
```

TRUE

6367

FALSE TRUE

8 6750

5114

```
[9]: # get TEs in common between Stellarscope and SoloTE
```

```
stellarscopeTEs_celltype <- Features(objTE_stellarscope_celltype)
```

```
table(stellarscopeTEs_celltype %in% conversionTable$stellarscopeID)
```

```
soloTEs_celltype <- Features(objTE_stellarscope_celltype)
```

```
# remove "SoloTE" from the name of the TEs and substitute "/" and "_" with "-"
```

```
soloTEs_celltype <- gsub("\\|", "-", soloTEs_celltype)
```

```
soloTEs_celltype <- gsub("\\_", "-", soloTEs_celltype)
```

```
soloTEs_celltype <- gsub("\\?", "", soloTEs_celltype)
```

TRUE

6235

```
[10]: # get TEs in common between Stellarscope and STARsolo
```

```
stellarscopeTEs <- Features(objTE_stellarscope)
```

```
table(stellarscopeTEs %in% conversionTable$stellarscopeID)
```

```
STARsoloTEs <- Features(objTE_STARsolo)
```

```
table(STARsoloTEs %in% conversionTable$stellarscopeID)
```

```
TEsInStellarscope_STARsolo <- intersect(  
  ↪conversionTable$locusID[conversionTable$stellarscopeID %in% stellarscopeTEs],  
  ↪conversionTable$locusID[conversionTable$stellarscopeID  
  ↪%in% STARsoloTEs] )
```

```
length(TEsInStellarscope_STARsolo)
```

TRUE

6367

TRUE

4955

3970

```
[11]: STARsoloTEs <- Features(objTE_STARsolo)
table(STARsoloTEs %in% conversionTable$stellarscopeID)

TEsInAll <- intersect( TEsInStellarscope_SoloTE,
                       conversionTable$locusID[conversionTable$stellarscopeID %in% STARsoloTEs] )

length(TEsInAll)
```

TRUE

4955

3420

```
[12]: commonCellsAll <- intersect(intersect(Cells(objTE_stellarscope),
      ↪ Cells(objTE_soloTE)), Cells(objTE_STARsolo))
length(commonCellsAll)
```

1056

```
[13]: ## locus id of common TEs
tes_soloTE_conv <- conversionTable$locusID[match(Features(objTE_soloTE),
      ↪ conversionTable$soloteID)]
tes_stellarscope_conv <-
      ↪ conversionTable$locusID[match(Features(objTE_stellarscope),
      ↪ conversionTable$stellarscopeID)]
tes_starsolo_conv <- conversionTable$locusID[match(Features(objTE_STARsolo),
      ↪ conversionTable$stellarscopeID)]

commonTEs_conv <- intersect(intersect(tes_soloTE_conv, tes_stellarscope_conv),
      ↪ tes_starsolo_conv)
commonTEs_conv <- commonTEs_conv[!is.na(commonTEs_conv)]
length(commonTEs_conv)
```

3420

```
[14]: # ALL common TEs
# revert to original name
common_soloTE <- conversionTable$soloteID[match(commonTEs_conv,
      ↪ conversionTable$locusID)]
common_stellarscope <- conversionTable$stellarscopeID[match(commonTEs_conv,
      ↪ conversionTable$locusID)]
```

```

common_STARsolo <- conversionTable$stellarscopeID[match(commonTEs_conv,
  ↪conversionTable$locusID)]

# extract matrix of common TEs
stellarscopeMatSub <- GetAssayData(objTE_stellarscope[common_stellarscope,
  ↪commonCellsAll], layer = "data")
setdiff(common_stellarscope, rownames(stellarscopeMatSub))
stellarscopeMatSub <- stellarscopeMatSub[common_stellarscope, commonCellsAll] #
  ↪to reorder

STARsoloMatSub <- GetAssayData(objTE_STARsolo[common_stellarscope,
  ↪commonCellsAll], layer = "data")
STARsoloMatSub <- STARsoloMatSub[common_stellarscope, commonCellsAll] # to
  ↪reorder

soloTEMatSub <- GetAssayData(objTE_soloTE[common_soloTE, commonCellsAll], layer
  ↪= "data")
soloTEMatSub <- soloTEMatSub[common_soloTE, commonCellsAll] # to reorder

identical(colnames(stellarscopeMatSub), colnames(soloTEMatSub))
identical(colnames(STARsoloMatSub), colnames(soloTEMatSub))

matSubs <- list()
matSubs[["Stellarscope"]] <- stellarscopeMatSub
matSubs[["SoloTE"]] <- soloTEMatSub
matSubs[["STARsolo"]] <- STARsoloMatSub

gc()

```

TRUE

TRUE

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	27162562	1450.7	56799402	3033.5	36043506	1925.0
	Vcells	240475880	1834.7	451524231	3444.9	450565477	3437.6

```

[15]: # matrices with common cells and TEs
mat_stellarscope <- as.matrix(stellarscopeMatSub)

mat_soloTE <- as.matrix(soloTEMatSub)

mat_STARsolo <- as.matrix(STARsoloMatSub)

gc()

```

A matrix: 2×6 of type dbl		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
	Ncells	27162929	1450.7	56799402	3033.5	36043506	1925.0
	Vcells	251311089	1917.4	451524231	3444.9	450565477	3437.6

2 Correlations

```
[16]: avgExprCommonTEs <- rowMeans(mat_stellarscope)
# compute average expression in the cells where each TE is expressed
sumExprCommonTEs <- rowSums(mat_stellarscope)
nCellsExprTEs <- rowSums(mat_stellarscope>0)
avgExprPerExprCell <- sumExprCommonTEs/nCellsExprTEs

TEcorrsCommonTEs <- list()

# compute spearman correlation of stellarscope and soloTE counts
TEcorrsCommonTEs[["Stellarscope_SoloTE"]] <- sapply(1:nrow(mat_stellarscope),
  FUN = function(i){
    cor(mat_stellarscope[i,], mat_soloTE[i,], method="spearman")
  })

# compute spearman correlation of stellarscope and STARsolo counts
TEcorrsCommonTEs[["Stellarscope_STARsolo"]] <- sapply(1:nrow(mat_stellarscope),
  FUN = function(i){
    cor(mat_stellarscope[i,], mat_STARsolo[i,], method="spearman")
  })

# compute spearman correlation of stellarscope and STARsolo counts
TEcorrsCommonTEs[["STARsolo_SoloTE"]] <- sapply(1:nrow(mat_STARsolo), FUN =
  function(i){
    cor(mat_STARsolo[i,], mat_soloTE[i,], method="spearman")
  })
```

```
[17]: summary(TEcorrsCommonTEs[["Stellarscope_SoloTE"]])
summary(TEcorrsCommonTEs[["Stellarscope_STARsolo"]])
summary(TEcorrsCommonTEs[["STARsolo_SoloTE"]])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.00494	0.95122	0.97601	0.95924	0.99022	1.00000

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.08309	0.92177	0.95619	0.92950	0.97577	0.99999

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.02319	0.89744	0.94411	0.90981	0.96982	0.99999

3 Checkpoint

```
[4]: #save.image("workspaces/
      ↪afterCorrelations_10xPBMc_wSTAR_noPlatelet_StellarscopeV1.5.RData")
load("workspaces/afterCorrelations_10xPBMc_wSTAR_noPlatelet_StellarscopeV1.5.
      ↪RData")
rm(mat_stellarscope)
rm(mat_soloTE)
rm(mat_STARsolo)
gc()
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	27127601	1448.8	52106177	2782.8	27141585	1449.6
	Vcells	232020433	1770.2	352312863	2688.0	242946700	1853.6

4 Post

4.1 Correlations

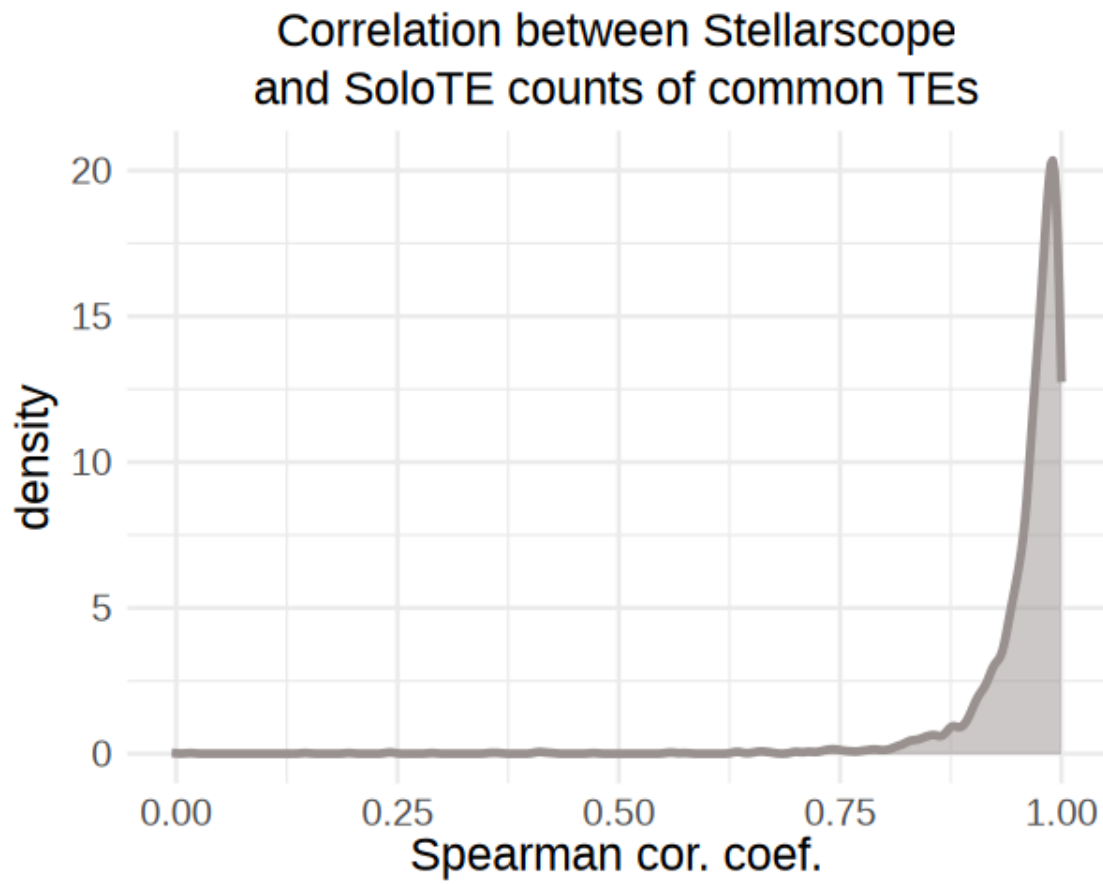
```
[4]: options(repr.plot.width=5, repr.plot.height=4)

for(combo in names(TEcorrsCommonTEs)){

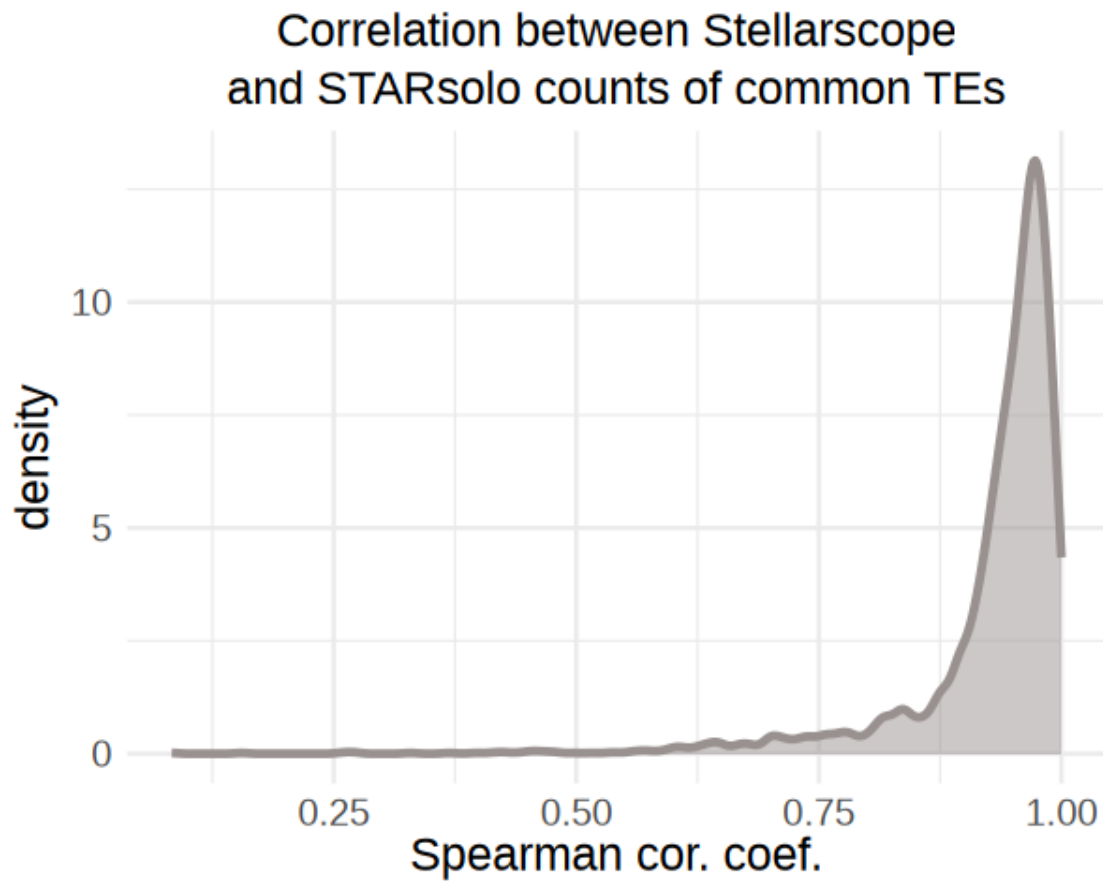
  tools <- unlist(strsplit(combo, "_"))
  show(
    ggplot() + aes(x=TEcorrsCommonTEs[[combo]]) +
      geom_density(color="#99918F", fill=alpha("#99918F",0.
↪5), linewidth = 1) +
      ggtitle(paste0("Correlation between ", tools[1]),
        subtitle = paste0("and ", tools[2], " counts of_
↪common TEs")) +
      xlab("Spearman cor. coef.") +
      theme_minimal() +
      theme(text=element_text(size=14.5),
        plot.title = element_text(size=14.5, hjust=0.
↪5),
        plot.subtitle = element_text(size=14.5, hjust=0.
↪5))
  )

  ggsave(paste0("figures_",dataset_id,"/
↪spearmanCorrelation_density_commonTEs_",combo,".pdf"), device = "pdf")
}
```

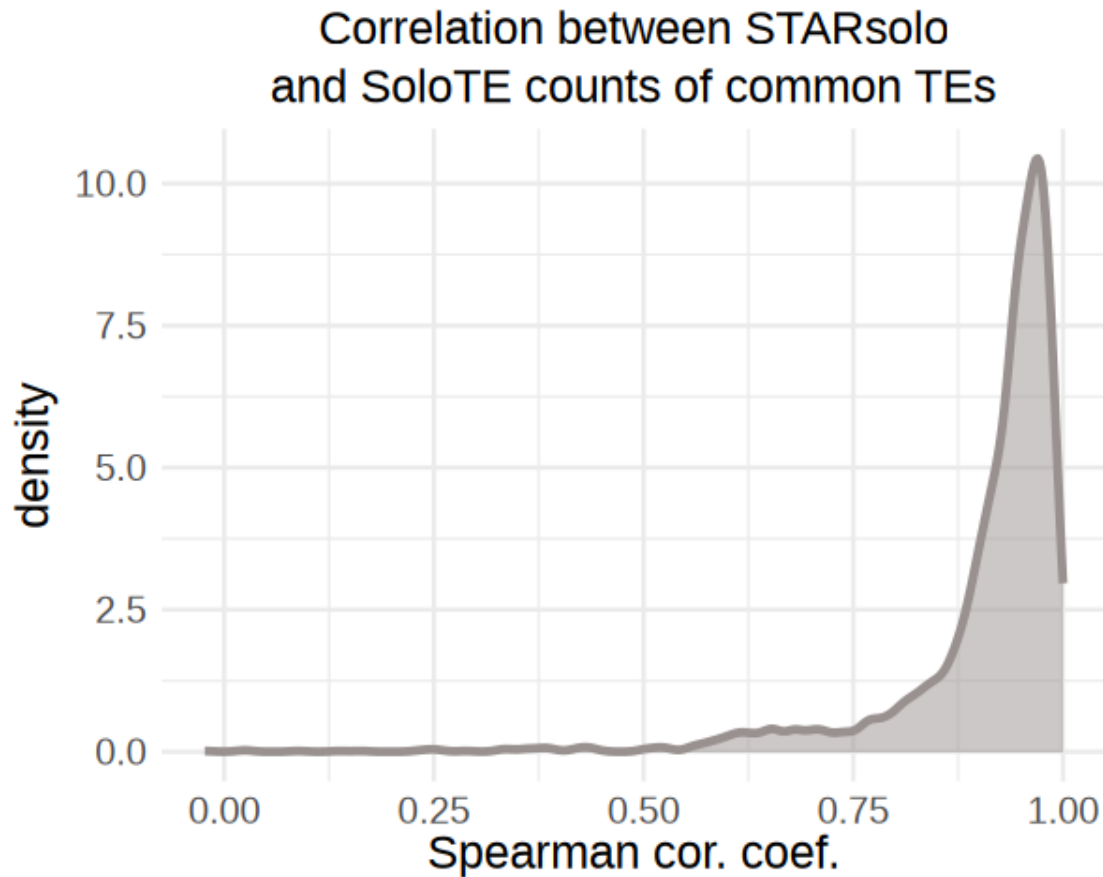
Saving 6.67 x 6.67 in image



Saving 6.67 x 6.67 in image



Saving 6.67 x 6.67 in image



```
[5]: options(repr.plot.width=5, repr.plot.height=4)

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))

  df <- cbind(avgExprPerExprCell, TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("avgExprPerExprCell", "TEcorrsCommonTEs")

  cat("Correlation between correlations and avg expression (", combo,") : ",
      cor(avgExprPerExprCell, TEcorrsCommonTEs[[combo]], method="spearman"), "\n")

  show(ggplot(df, aes(x=TEcorrsCommonTEs, y=avgExprPerExprCell)) +
    ggtitle(combo) +
    geom_point(alpha= 0.3))

  # divide by expr quartiles
  df$quartile <- "4"
```

```

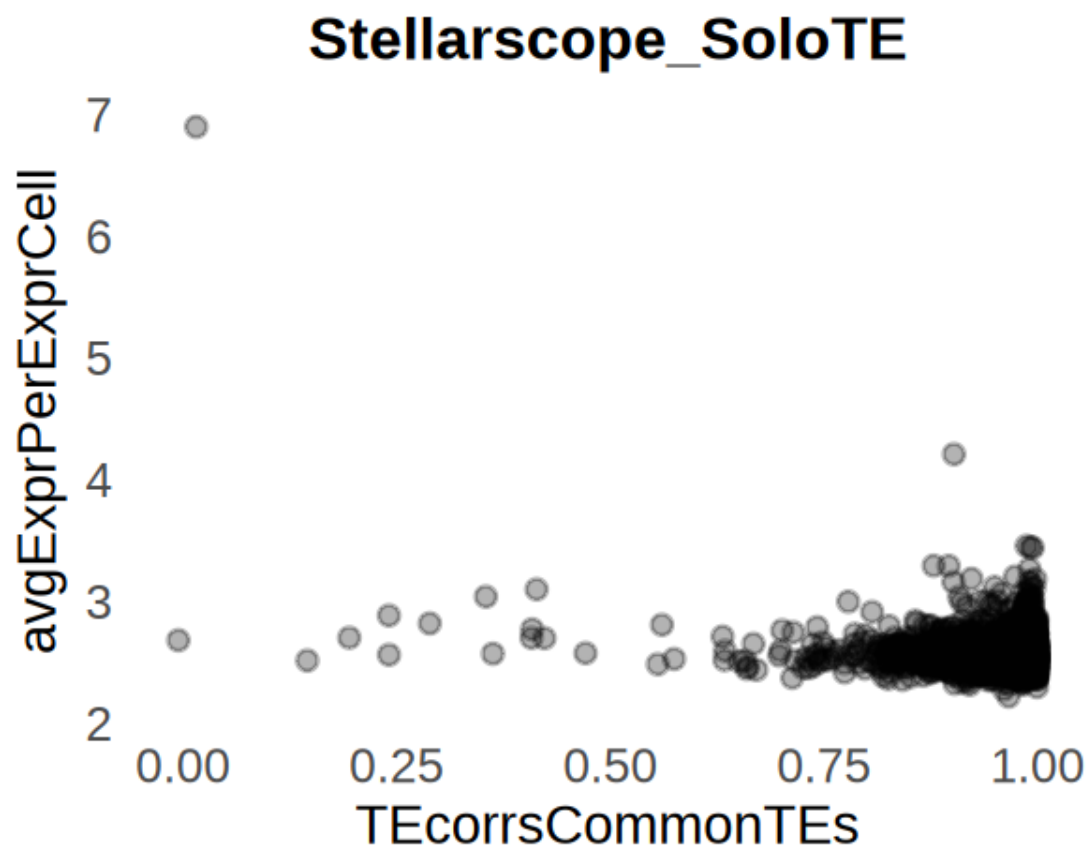
df$quartile[df$avgExprPerExprCell < quantile(df$avgExprPerExprCell, 0.75, na.
↪rm=TRUE)] <- "3"
df$quartile[df$avgExprPerExprCell < quantile(df$avgExprPerExprCell, 0.5, na.
↪rm=TRUE)] <- "2"
df$quartile[df$avgExprPerExprCell < quantile(df$avgExprPerExprCell, 0.25, na.
↪rm=TRUE)] <- "1"

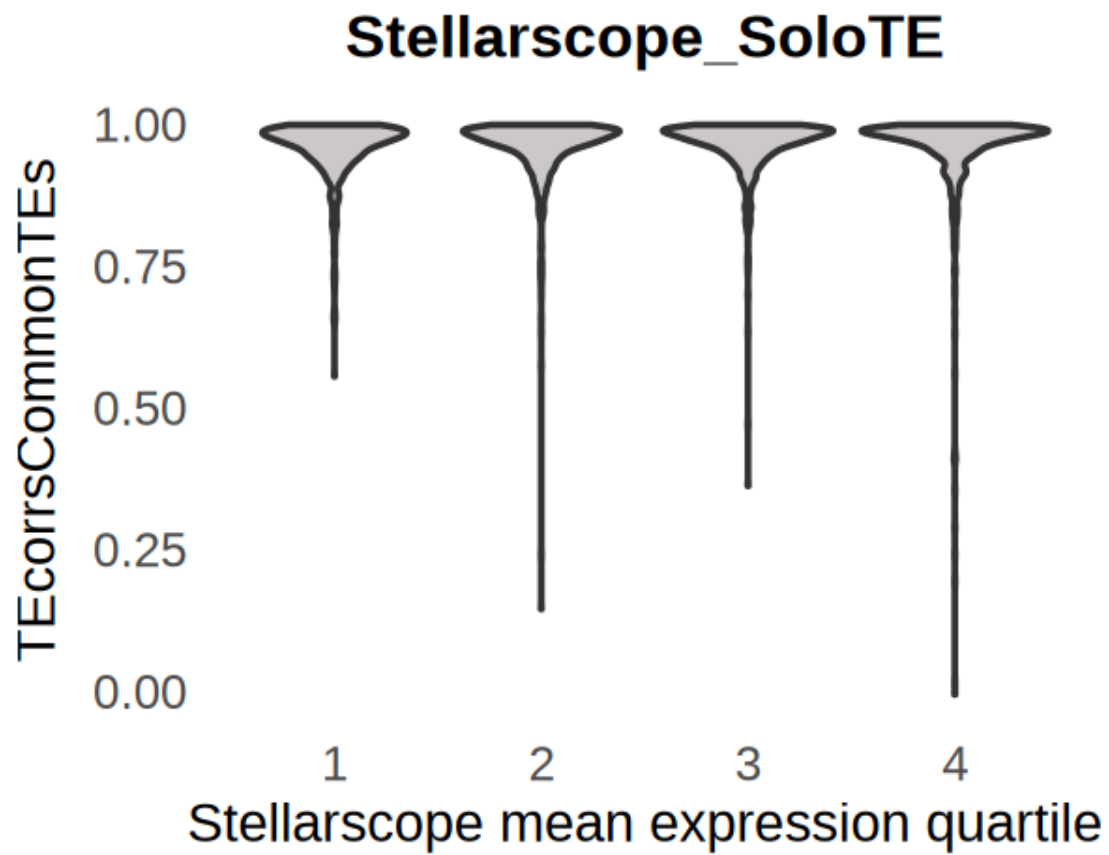
show(ggplot(df, aes(x=quartile, y=TEcorrsCommonTEs)) + ggtitle(combo) +
  geom_violin(fill=alpha("#99918F",0.5)) +
  theme(text=element_text(size=15)) +
  xlab("Stellarscope mean expression quartile") )

# divide by correlation quartiles
df$quartile <- "4"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
↪rm=TRUE)] <- "3"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
↪rm=TRUE)] <- "2"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
↪rm=TRUE)] <- "1"
show(ggplot(df, aes(x=quartile, y=avgExprPerExprCell)) +
  geom_violin(fill=alpha("#99918F",0.5)) + ggtitle(combo) +
  theme(text=element_text(size=15)) +
  scale_y_continuous(transform = "log") +
  xlab("Correlation quartile") )
}

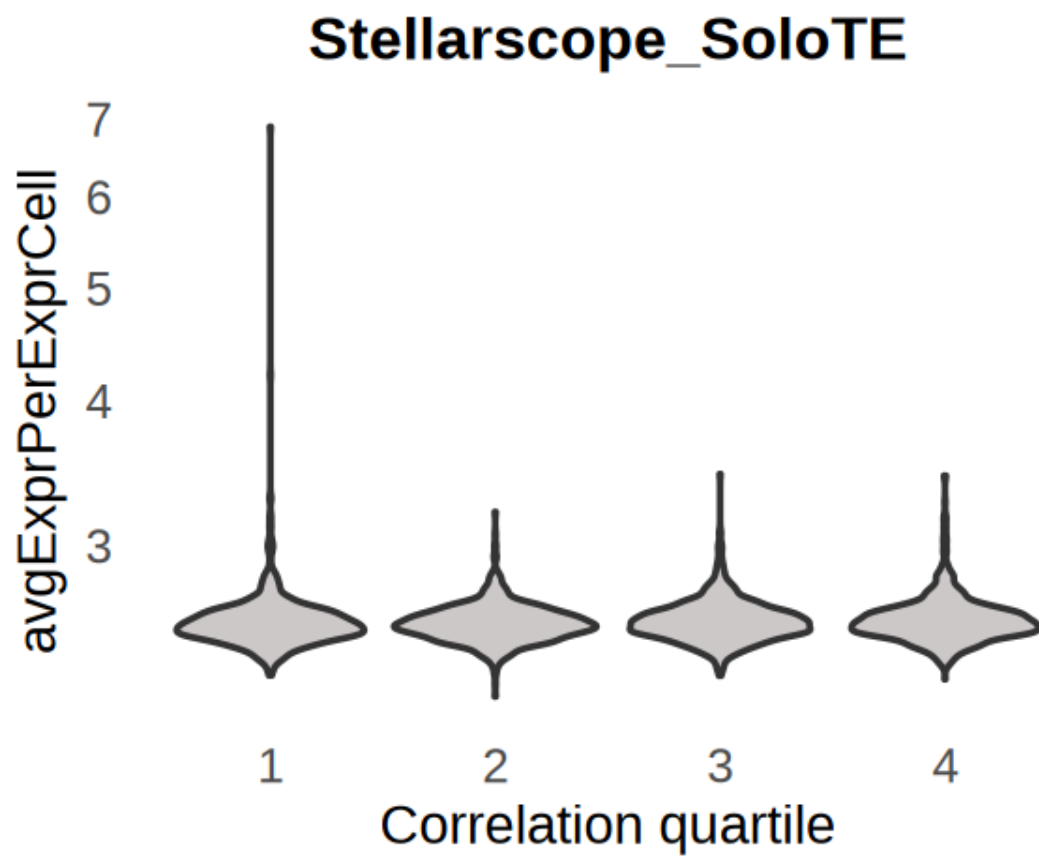
```

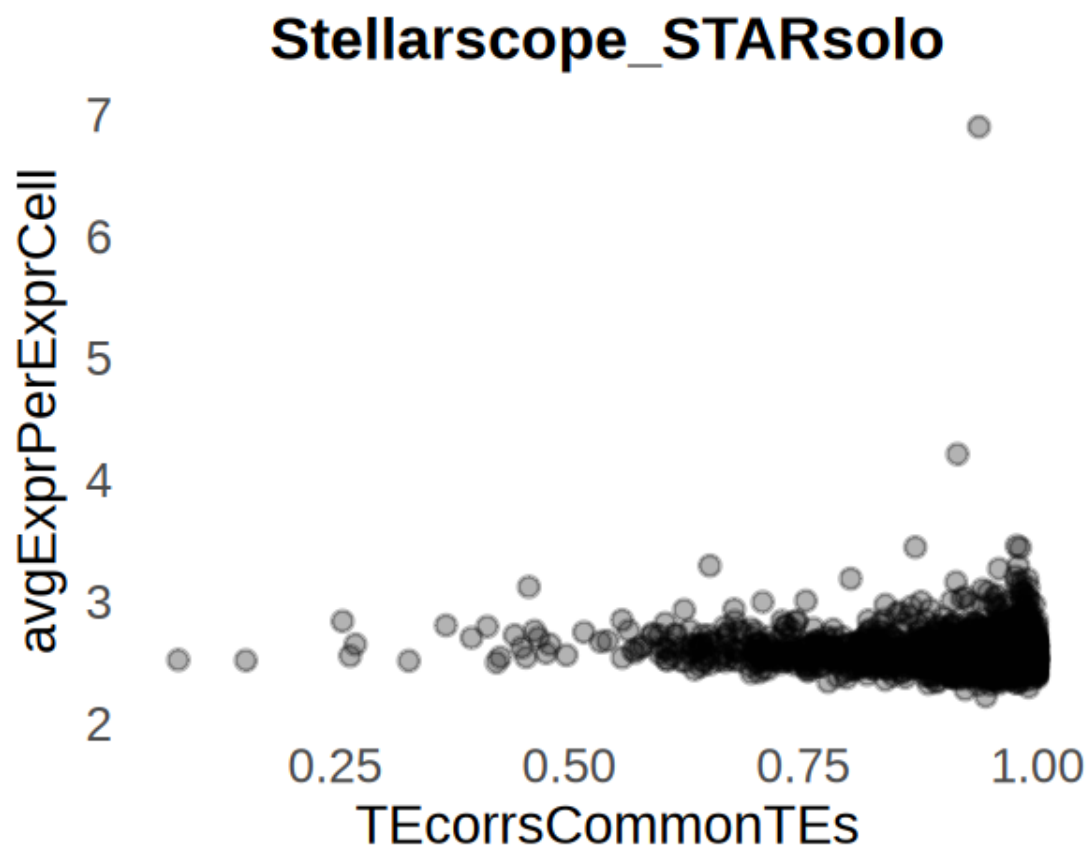
Correlation between correlations and avg expression (Stellarscope_SoloTE) :
0.05800211

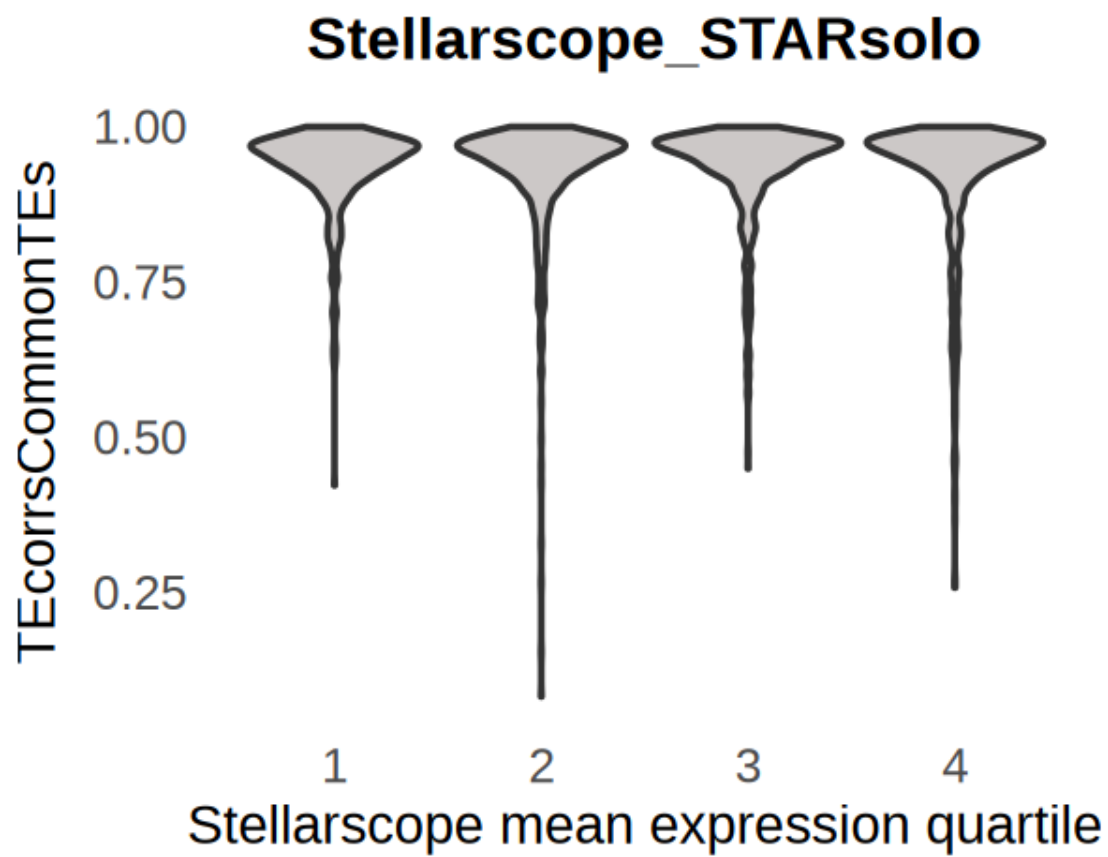




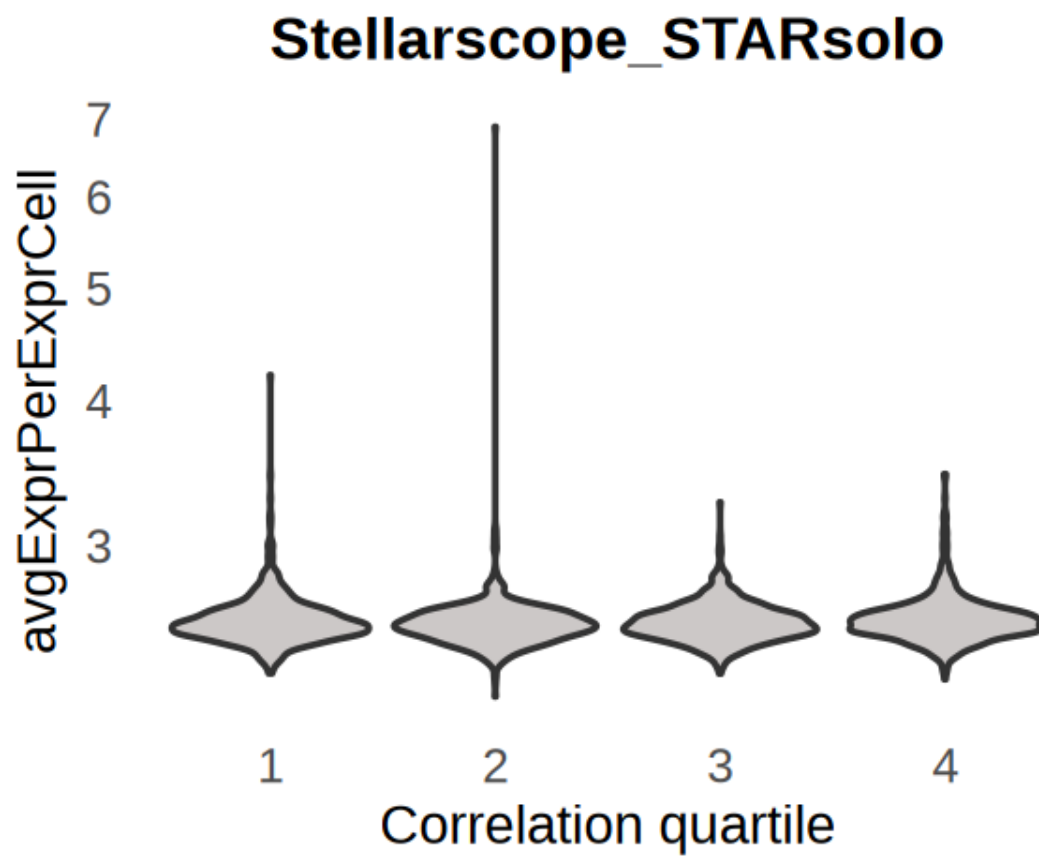
Correlation between correlations and avg expression (Stellarscope_STARsolo) :
0.02986781

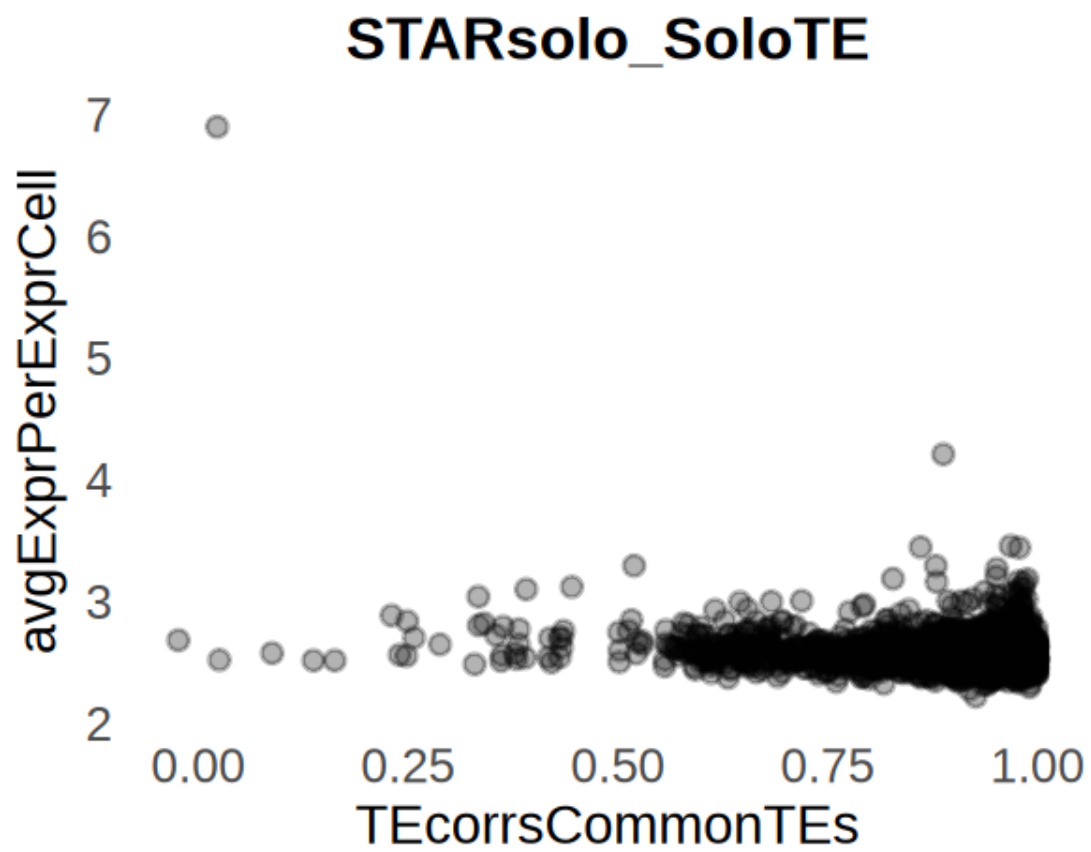


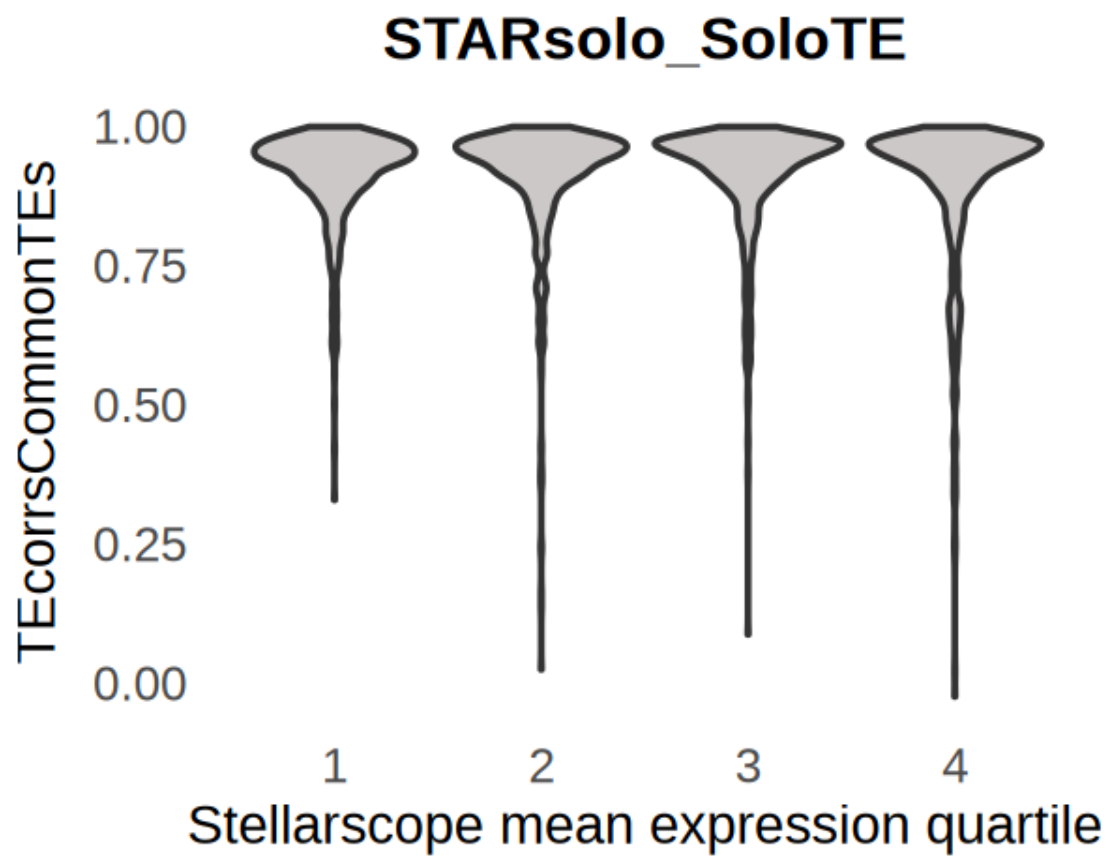


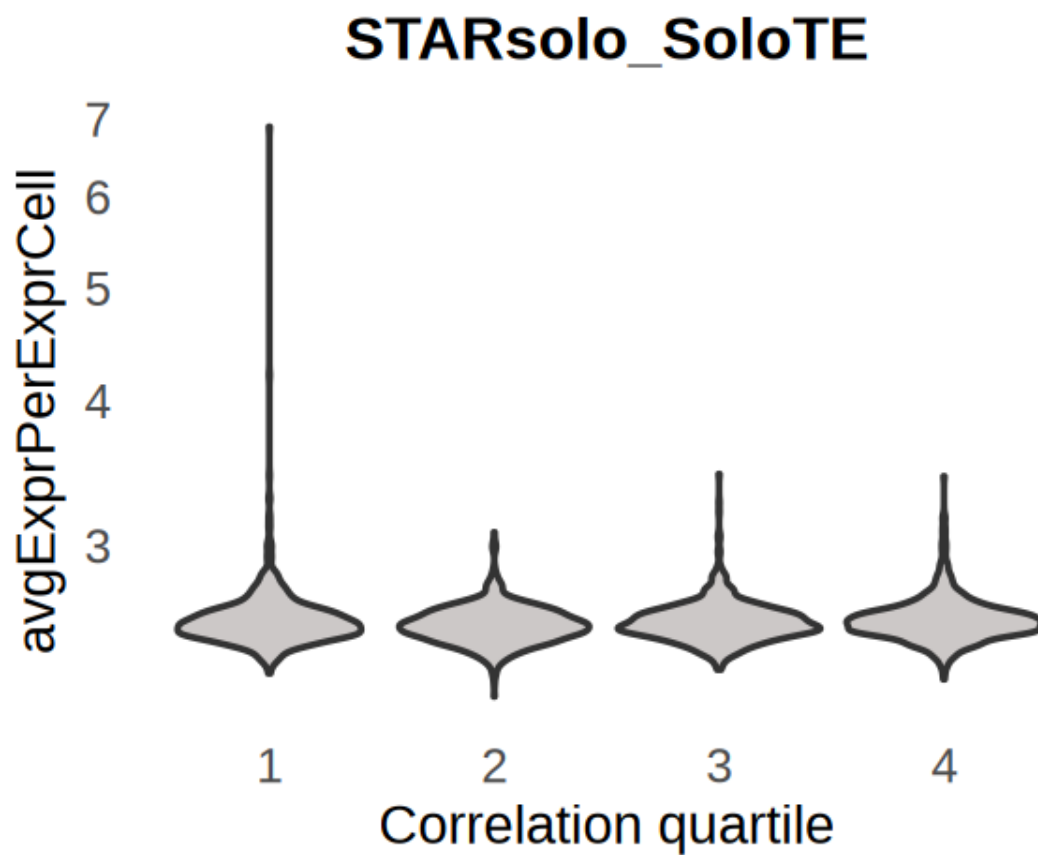


Correlation between correlations and avg expression (STARsolo_SoloTE) :
0.02616544









4.2 Correlation by age

```
[6]: # define age breaks for bins
breaks <- c(0, 2, 5, 10, 15, 25, 40, 60, Inf)

conversionTable$age_bin <- cut(
  conversionTable$mya,
  breaks = breaks,
  labels = c("0-2", "2-5", "5-10", "10-15", "15-25", "25-40", "40-60", "60+"),
  right = FALSE,
  include.lowest = TRUE
)

table(conversionTable$age_bin)
```

0-2	2-5	5-10	10-15	15-25	25-40	40-60	60+
13303	39956	248223	632189	1102057	1438502	1028605	26428

```

[7]: options(repr.plot.width=10, repr.plot.height=6)
library(ggbeeswarm)

dfsCorrelationAge <- list()

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))

  df <- cbind(rownames(matSubs[[tools[1]]]), TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("V1", "TEcorrsCommonTEs")

  df$ageClass <- conversionTable$ageClass[match(df$V1,
↪conversionTable$stellarscopeID)]
  df$age_bin <- conversionTable$age_bin[match(df$V1,
↪conversionTable$stellarscopeID)]
  df$mya <- conversionTable$mya[match(df$V1, conversionTable$stellarscopeID)]
  df$TEcorrsCommonTEs <- as.numeric(df$TEcorrsCommonTEs)
  cor(df$mya, df$TEcorrsCommonTEs, method="spearman")

  show( ggplot(df, aes(x=age_bin, y=TEcorrsCommonTEs, fill=age_bin)) +
        geom_violin(alpha = 0.8) +
        stat_summary(fun.data = function(x) data.frame(y = max(x), label =
↪length(x)),
        geom = "text", vjust = -0.5, size = 5, color = "black") +
        theme_paper() +
        #scale_fill_manual(values=colorAge) +
        guides(fill = "none") +
        theme(text=element_text(size=18)) +
        xlab("Age bins (mya)") +
        ggtitle(combo)
      )

  show( ggplot(df, aes(x=age_bin, y=TEcorrsCommonTEs, color=age_bin)) +
        geom_beeswarm(alpha = 0.7, cex = 0.2, size=0.5, width=0.5) +
        #geom_boxplot(outlier.shape = NA, width = 0.3, color = "gray40",
↪fill=NA, alpha = 0.3) +
        stat_summary(
          fun = median,
          geom = "crossbar",
          width = 0.3,
          color = "grey30",
          linewidth = 0.5
        ) +
        stat_summary(fun.data = function(x) data.frame(y = max(x), label =
↪length(x)),

```

```

        geom = "text", vjust = -0.5, size = 5, color = "black") +
    theme_paper() +
    #scale_fill_manual(values=colorAge) +
    guides(color = "none") +
    theme(text=element_text(size=18)) +
    xlab("Age bins (mya)") +
    ylab("Spearman cor. coef.") +
    ggtitle(combo)
)
ggsave(paste0("figures_",dataset_id,"/correlation_byAge_Bin_",combo,".pdf"),
width=8, height=6)

show( ggplot(df, aes(x=ageClass, y=TEcorrsCommonTEs, fill=ageClass)) +
    geom_violin(alpha = 0.5) +
    stat_summary(fun.data = function(x) data.frame(y = max(x), label =
length(x)),
        geom = "text", vjust = -0.5, size = 5, color = "black") +
    theme_pubclean() +
    scale_fill_manual(values=colorAge) +
    theme(text=element_text(size=18)) +
    xlab("Age class") +
    ggtitle(combo)
)

# divide by correlation quartiles
df$quartile <- "4"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
rm=TRUE)] <- "3"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
rm=TRUE)] <- "2"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
rm=TRUE)] <- "1"

show(ggplot(df, aes(x=quartile, y=mya)) +
    geom_violin(fill=alpha("#99918F",0.5)) +
    stat_summary(fun.data = function(x) data.frame(y = max(x), label =
length(x)),
        geom = "text", vjust = -0.5, size = 5) +
    theme_pubclean() +
    theme(text=element_text(size=18)) +
    scale_y_continuous(transform = "log") +
    xlab("Correlation quartile") +
    ggtitle(combo)
)

dfsCorrelationAge[[combo]] <- df

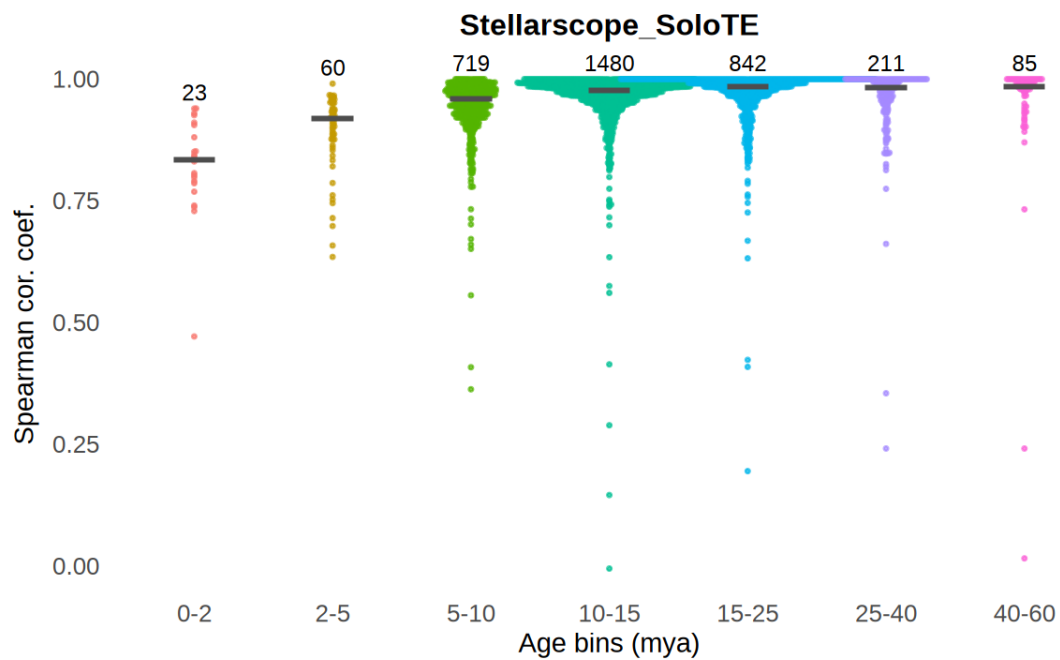
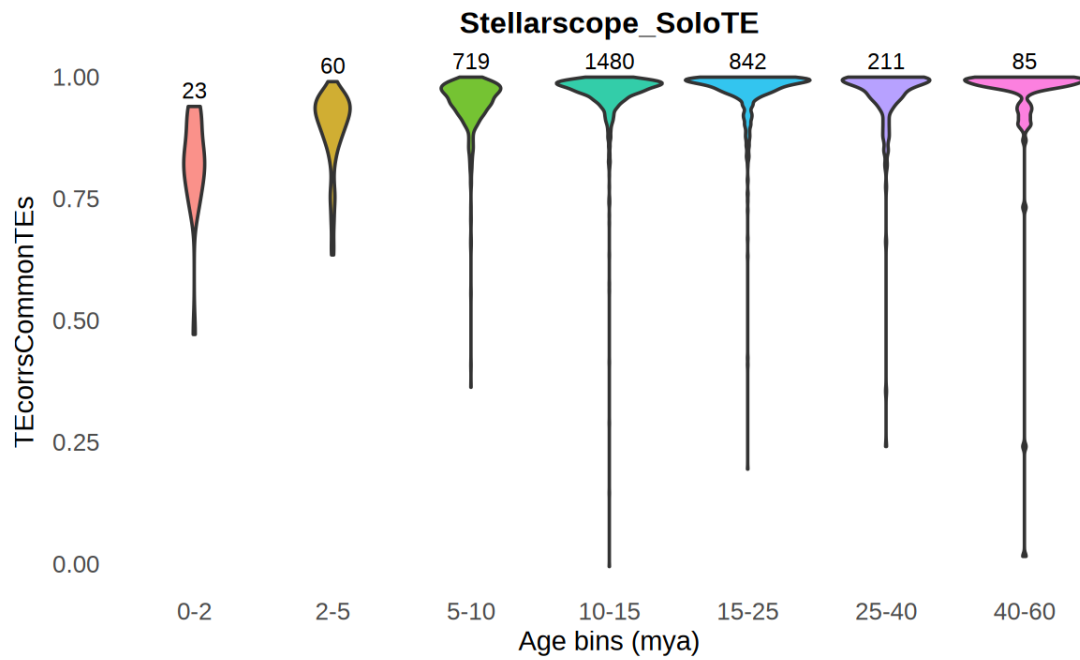
```

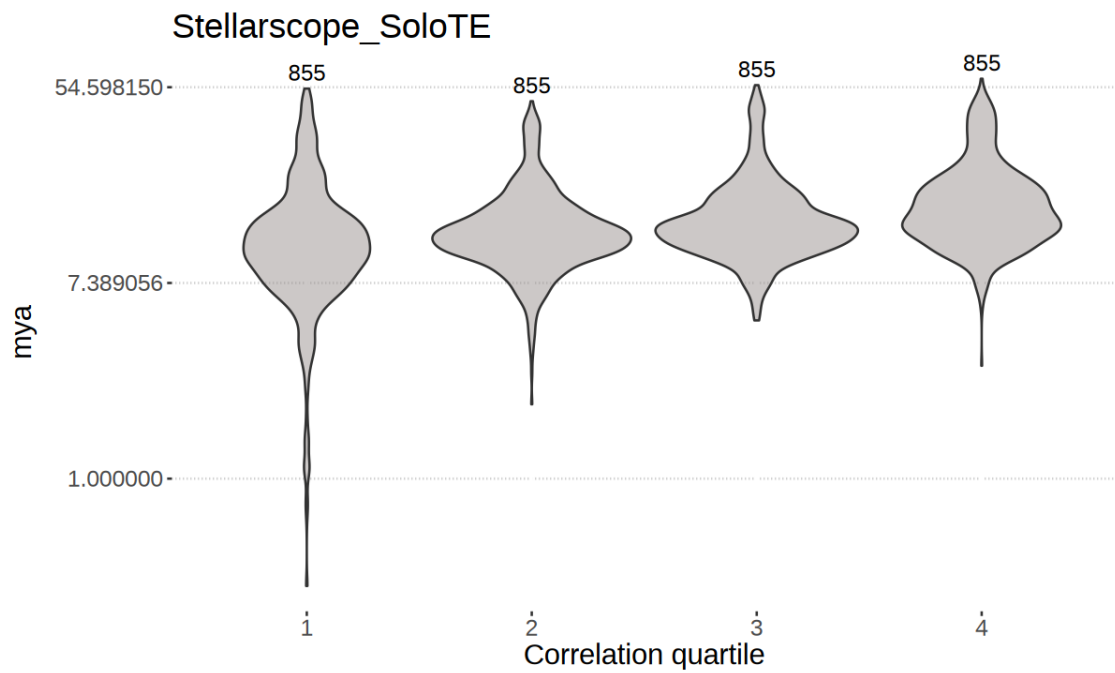


```
}
```

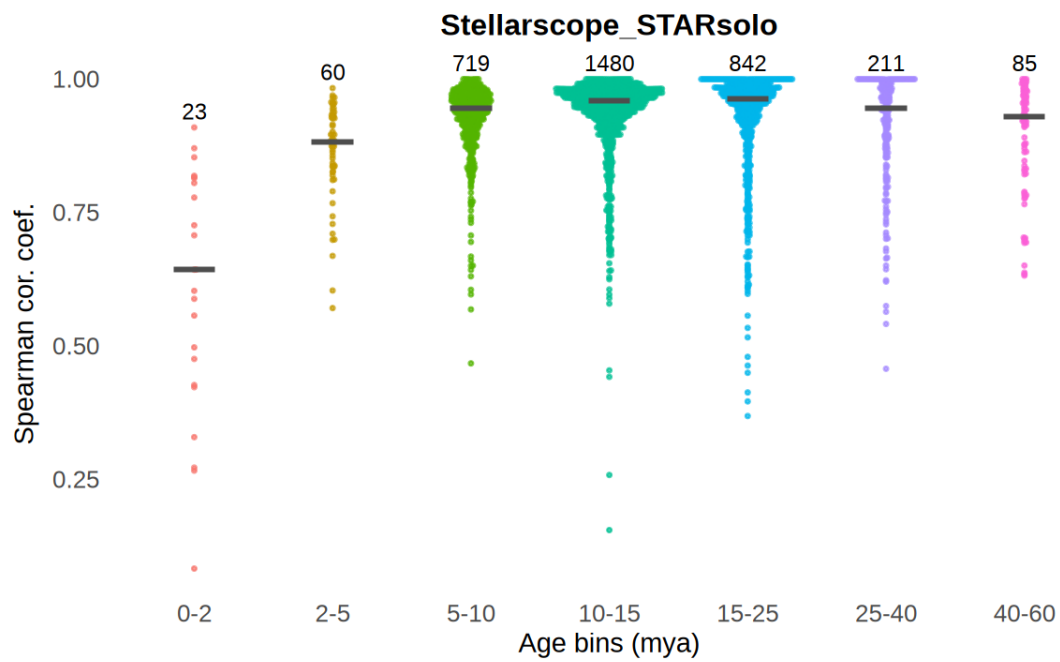
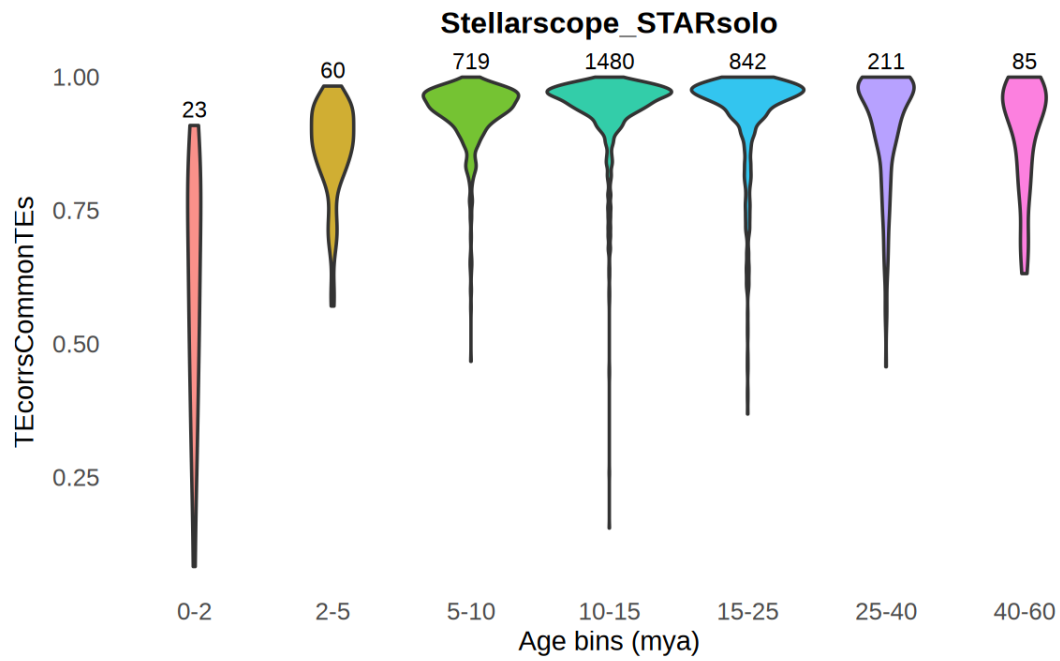
Warning message in `geom_beeswarm(alpha = 0.7, cex = 0.2, size = 0.5, width = 0.5)`:

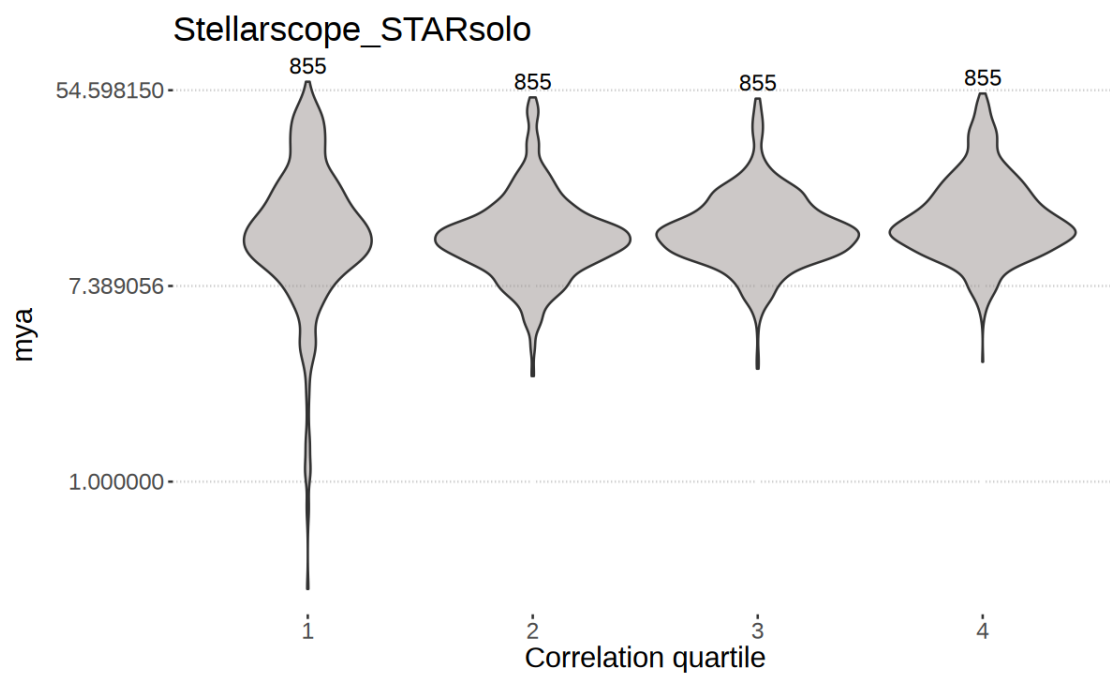
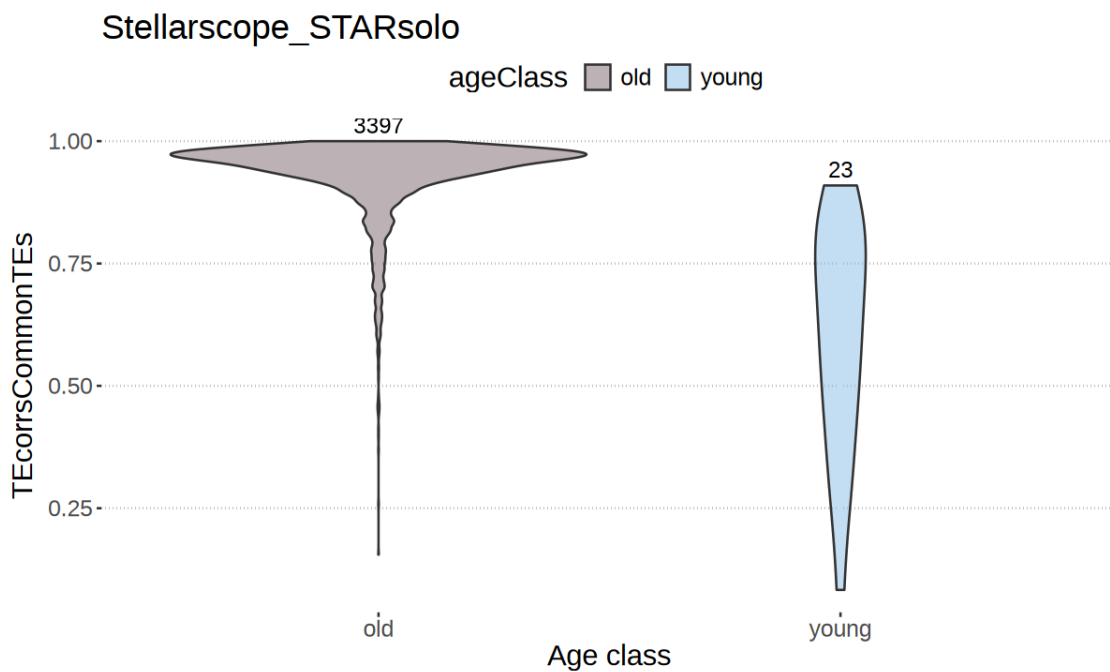
"Ignoring unknown parameters: ``width``"



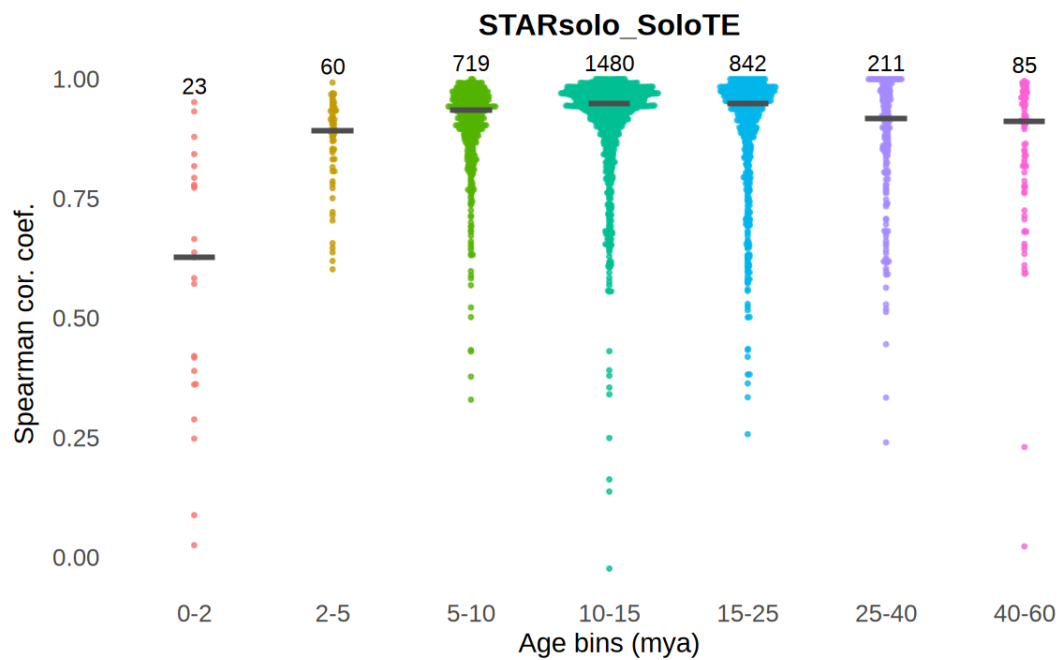
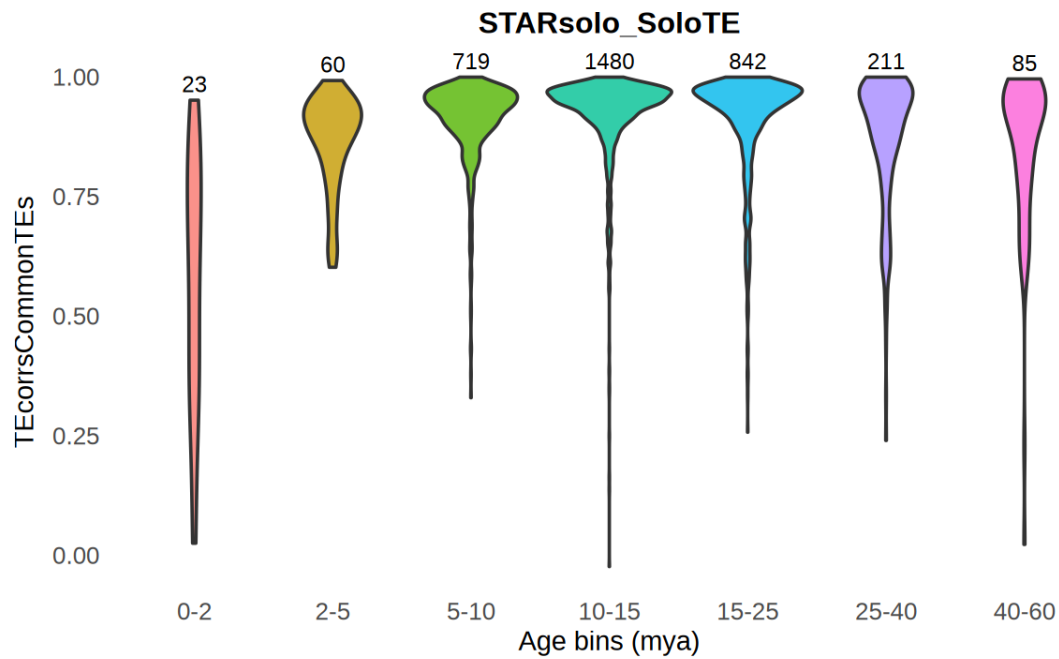


Warning message in `geom_beeswarm(alpha = 0.7, cex = 0.2, size = 0.5, width = 0.5)`:
 "Ignoring unknown parameters: `width`"





Warning message in geom_beeswarm(alpha = 0.7, cex = 0.2, size = 0.5, width = 0.5):
 "Ignoring unknown parameters: `width`"





```
[8]: options(repr.plot.width=12, repr.plot.height=5)

ggplot(dfsCorrelationAge[["Stellarscope_SoloTE"]], aes(x=mya)) +
  geom_density(alpha = 0.5) +
```

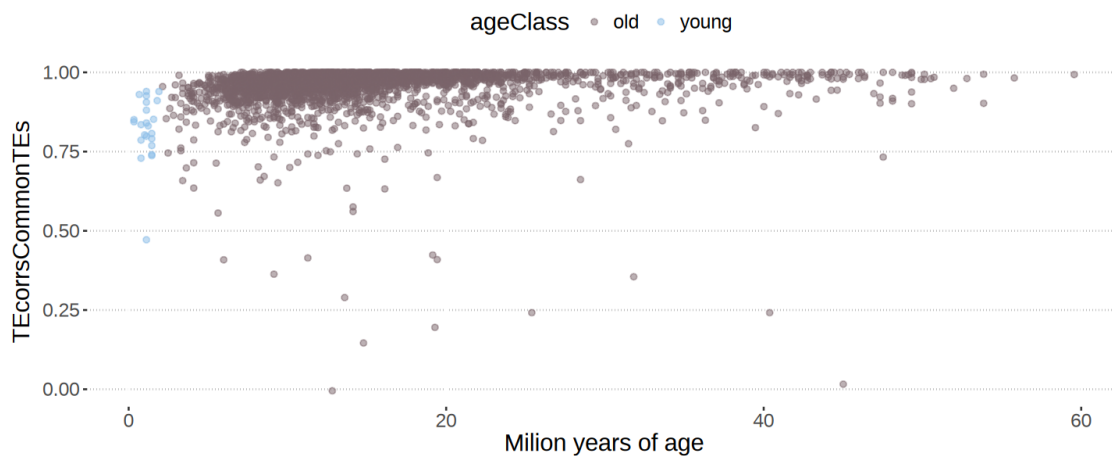
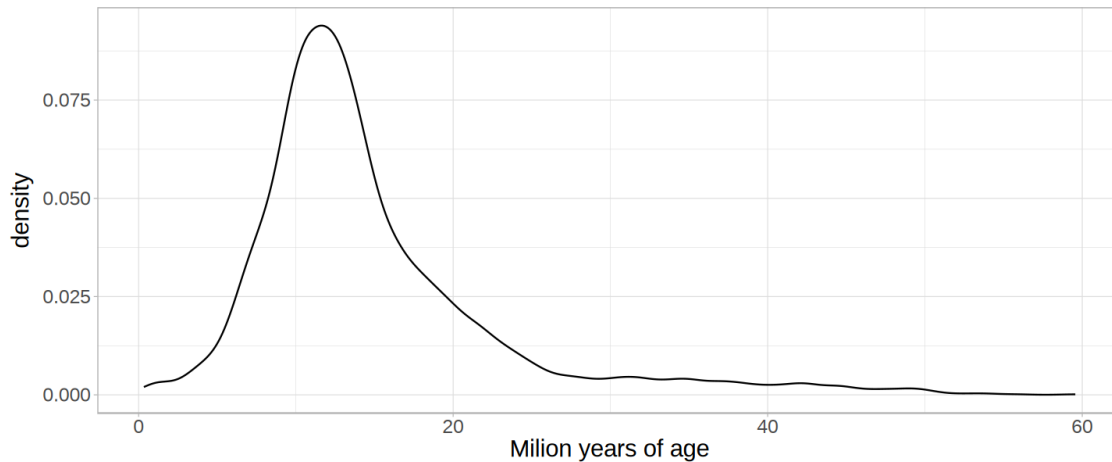
```

theme_light() +
theme(text=element_text(size=18)) +
xlab("Milion years of age")

ggplot(dfsCorrelationAge[["Stellarscope_SoloTE"]], aes(x=mya,
y=TEcorrsCommonTEs, color=ageClass)) +
  geom_point(alpha = 0.5, ) +
  scale_color_manual(values=colorAge) +
  theme_pubclean() +
  theme(text=element_text(size=18)) +
  xlab("Milion years of age")

# this is only for the commonly detected TEs
# the pattern with tes common to SoloTE and Stellarscope was a bit different

```

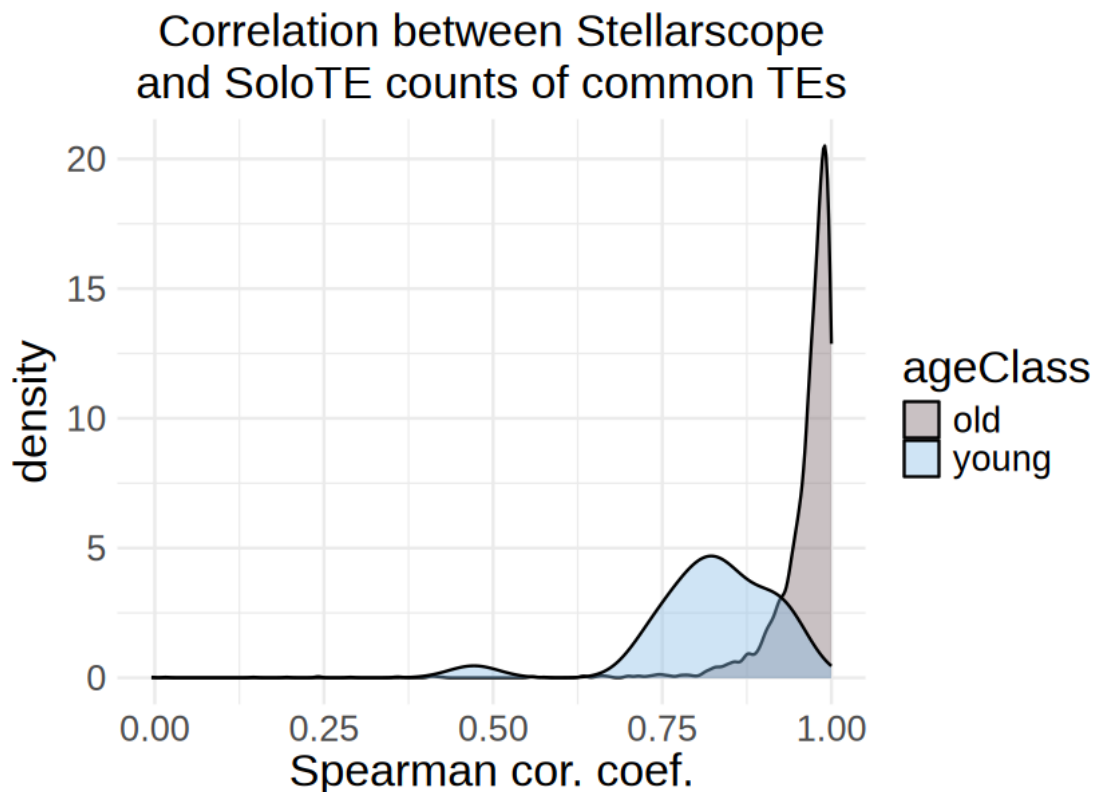


```
[24]: options(repr.plot.width=7, repr.plot.height=5)

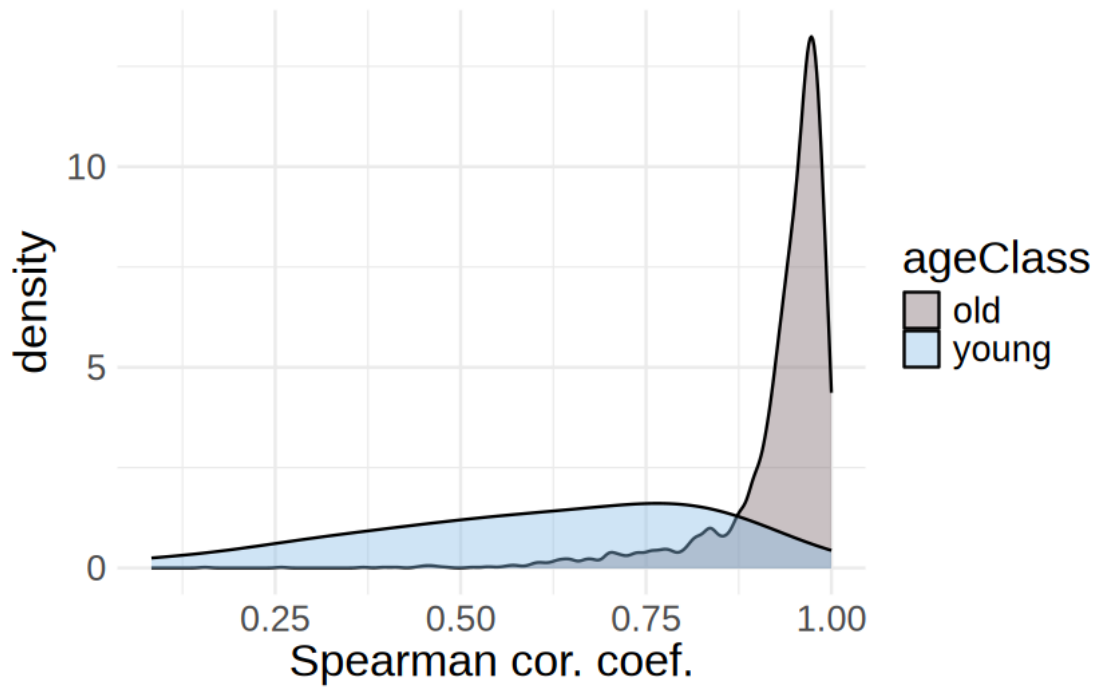
for(combo in names(dfsCorrelationAge)){

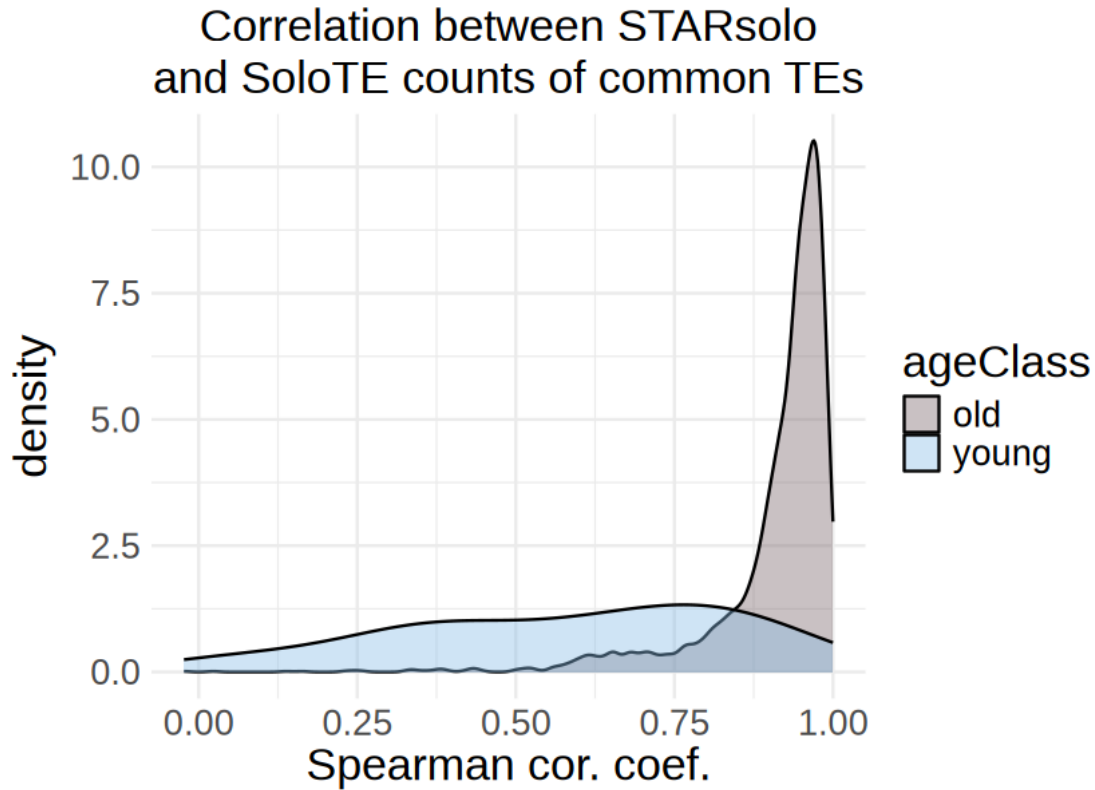
  tools <- unlist(strsplit(combo, "_"))

  show(ggplot(dfsCorrelationAge[[combo]]) + aes(x=TEcorrsCommonTEs,
    ↪fill=ageClass) +
    geom_density( linewidth = 0.5, alpha = 0.4 ) +
    ggtitle(paste0("Correlation between ", tools[1]),
    ↪subtitle = paste0("and ", tools[2], " counts of common_
    ↪TEs")) +
    xlab("Spearman cor. coef.") +
    scale_fill_manual(values=colorAge) +
    theme_minimal() +
    theme(text=element_text(size=20),
    plot.title = element_text(size=20, hjust=0.5),
    plot.subtitle = element_text(size=20, hjust=0.5))
  )
  ggsave(paste0("figures_", dataset_id, "/correlation_byAge_", combo, ".
    ↪pdf"), width=7, height=5)
}
```



Correlation between Starscope
and STARsolo counts of common TEs





4.3 Correlation by family

5 Statistics on n features

```
[29]: options(repr.plot.width=5, repr.plot.height=7)

nCountsSoloTE <- objTE_soloTE[,commonCellsAll]$nCount_RNA
nCountsStellarscope <- objTE_stellarscope[,commonCellsAll]$nCount_TE
nCountsSTARsolo <- objTE_STARsolo[,commonCellsAll]$nCount_TE

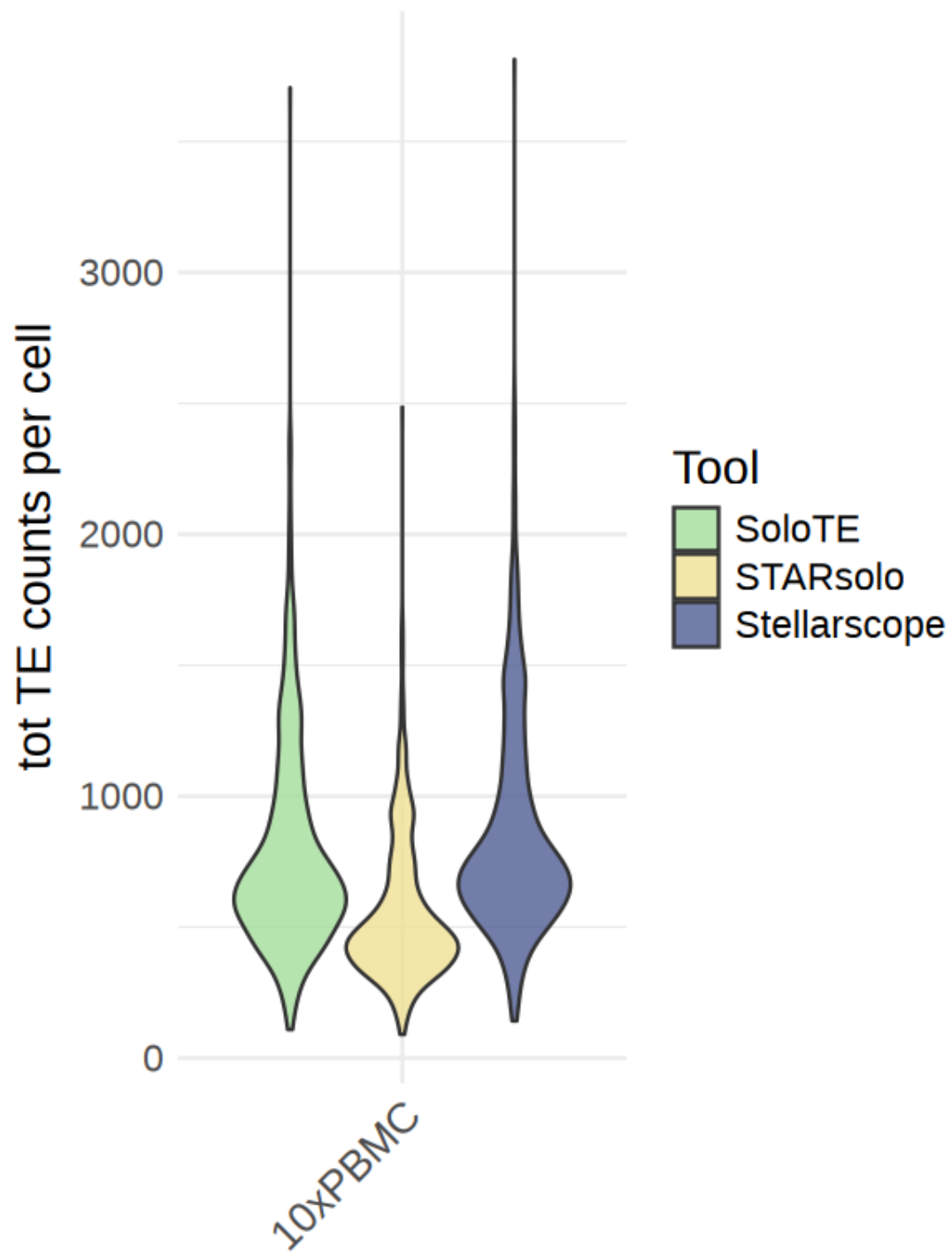
df <- as.data.frame(rbind(cbind(nCountsSoloTE,"SoloTE",␣
  ↪apply(strsplit(names(nCountsSoloTE), "_"),'[,1)),
                                cbind(nCountsStellarscope,"Stellarscope",␣
  ↪apply(strsplit(names(nCountsStellarscope), "_"),'[,1)),
                                cbind(nCountsSTARsolo,"STARsolo",␣
  ↪apply(strsplit(names(nCountsSTARsolo), "_"),'[,1)))))

colnames(df) <- c("nCounts_TEs","Tool","CB")
df$Sample <- dataset_id
df$nCounts_TEs <- as.numeric(df$nCounts_TEs)
```

```

ggplot(df, aes(x=Sample, fill=Tool, y=nCounts_TEs)) +
  geom_violin(scale = "width", alpha = 0.8, color="grey20") +
  scale_fill_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("tot TE counts per cell") +
  theme_minimal() +
  theme(text=element_text(size=17),
        plot.title = element_text(size=17, hjust=0.5),
        plot.subtitle = element_text(size=17, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/totCounts_SoloTEvsStellarscopevsSTARsolo.
↳pdf"), device = "pdf", width=5, height=7)

```



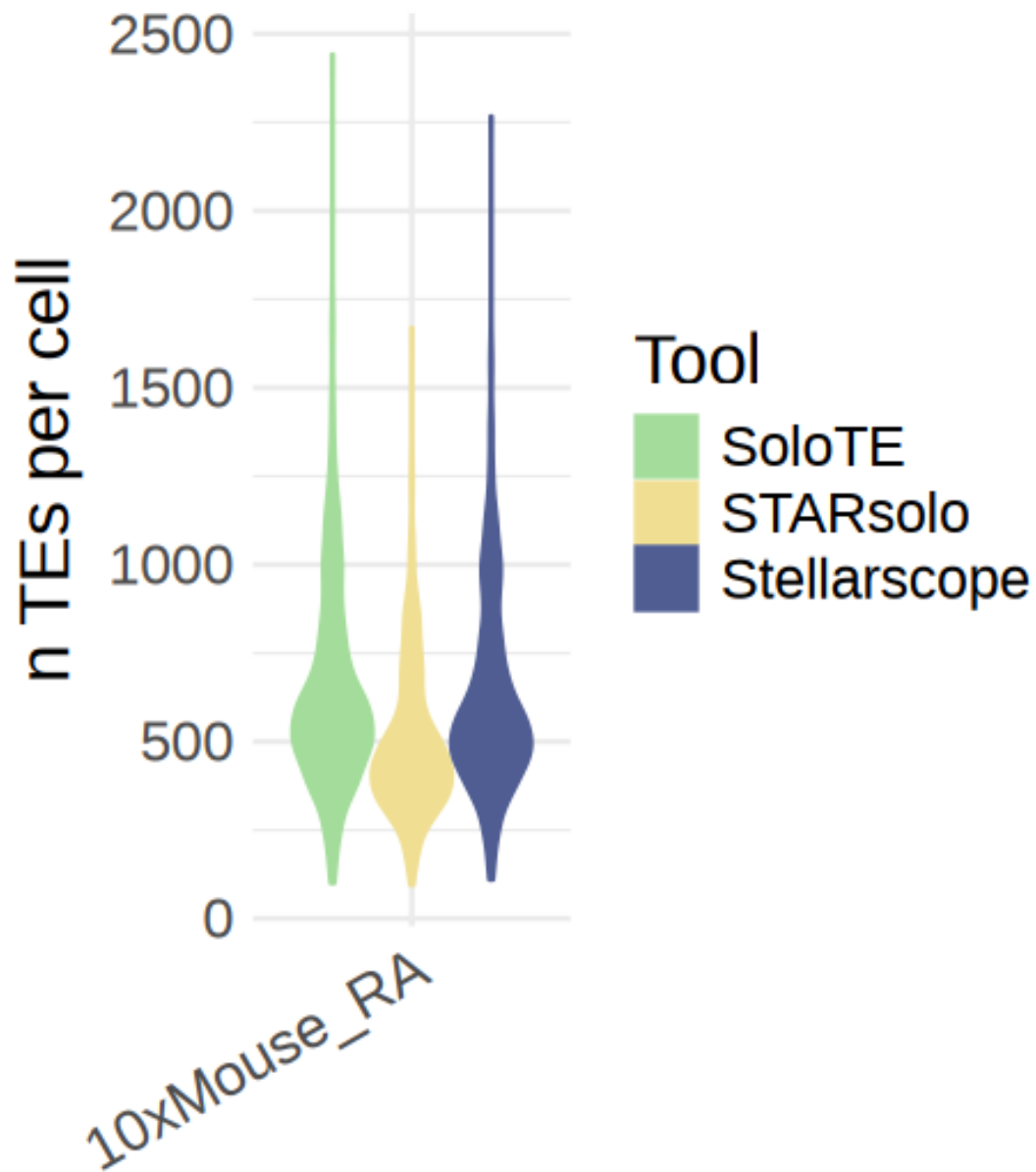
```
[30]: options(repr.plot.width=4, repr.plot.height=5)

nFeaturesSoloTE <- objTE_soloTE[,commonCellsAll]$nFeature_RNA
nFeaturesStellarscope <- objTE_stellarscope[,commonCellsAll]$nFeature_TE
nFeaturesSTARsolo <- objTE_STARsolo[,commonCellsAll]$nFeature_TE

df <- as.data.frame(rbind(cbind(nFeaturesSoloTE,"SoloTE",␣
  ↪apply(strsplit(names(nFeaturesSoloTE), "_"),'[,1)),
                        cbind(nFeaturesStellarscope,"Stellarscope",␣
  ↪apply(strsplit(names(nFeaturesStellarscope), "_"),'[,1)),
                        cbind(nFeaturesSTARsolo,"STARsolo",␣
  ↪apply(strsplit(names(nFeaturesSTARsolo), "_"),'[,1))))

colnames(df) <- c("nFeatures_TEs","Tool","Sample")
df$Sample <- "10xMouse_RA"
df$nFeatures_TEs <- as.numeric(df$nFeatures_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))

ggplot(df, aes(x=Sample, fill=Tool, color=Tool, y=nFeatures_TEs)) +
  geom_violin(scale = "width") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("n TEs per cell") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=30, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
  ↪totFeatures_SoloTEvsStellarscopevsSTARsolo_violin.pdf"), device = "pdf",␣
  ↪width=4, height=5)
```



```
[31]: objTE_stellarscope
      mat <- GetAssayData(objTE_stellarscope, layer="counts")
      sum(rowSums(mat > 0) >= 1)
```

An object of class Seurat

6367 features across 1056 samples within 1 assay
Active assay: RNA (6367 features, 4000 variable features)
3 layers present: counts, data, scale.data
2 dimensional reductions calculated: pca, umap
6367

```
[32]: # save lists of TEs expressed in each sample
nTEs_soloTE <- list()
TEs_soloTE <- list()
for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],
  ↪layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= 15)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= 15]
}
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
  ↪ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= 15)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= 15]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident],
  ↪layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= 15)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= 15]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE),
  ↪rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
             cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope),
  ↪rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
```

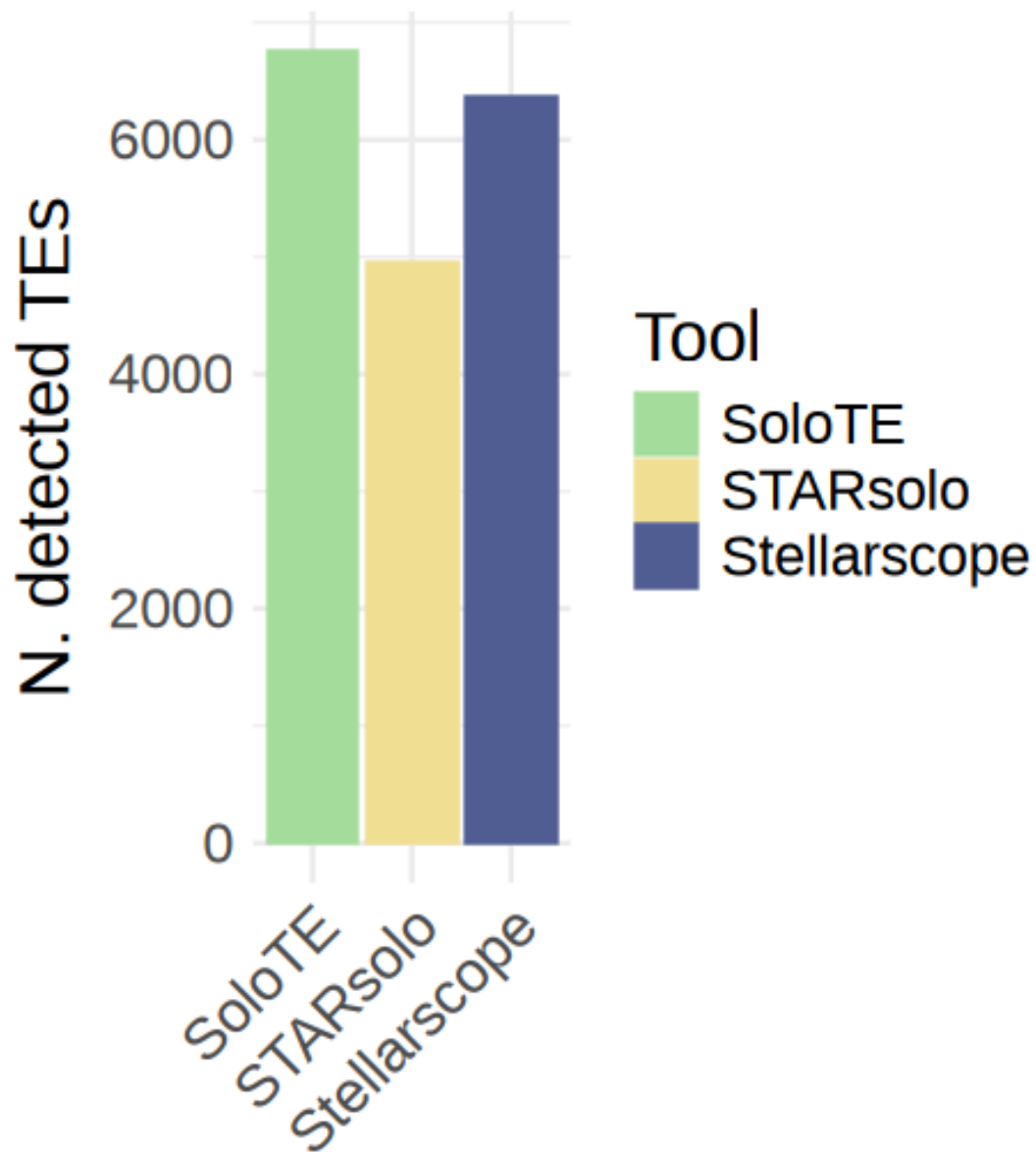
```

        cbind(unlist(nTEs_star), names(nTEs_star),
↪rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs", "Sample", "Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("N. detected TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot.pdf"), device = "pdf", width =
↪4, height=5)

```

		n_TEs	Sample	Tool
		<dbl>	<fct>	<chr>
A data.frame: 3 × 3	PBMC	6757	PBMC	SoloTE
	PBMC_pseudobulk	6366	PBMC_pseudobulk	Stellarscope
	STARsolo_TE_EM	4955	STARsolo_TE_EM	STARsolo



```
[33]: library(UpSetR)

options(repr.plot.width=10, repr.plot.height=5)
```

```

detectedList <- list(SoloTE = tes_soloTE_conv, Stellarscope = ☐
  ↪ tes_stellarscope_conv, STARsolo = tes_starsolo_conv)

UpSetR::upset(fromList(detectedList), nintersects = 30,
  sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
  text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
  order.by=c("freq"),
  point.size=3, mb.ratio = c(0.5, 0.5),
  set_size.show=F, set_size.scale_max=☐
  ↪ max(lengths(detectedList))+1000, #show.numbers = F,
  nsets=length(objList)+1,
  sets.x.label="N. detected loci",
  mainbar.y.label="Intersection size")
  ggsave(paste0("figures_",dataset_id,"/
  ↪ nTEs_SoloTEvsStellarscopevsSTARsolo_barplot.pdf"), device = "pdf", width =☐
  ↪ 10, height=5)

```

Warning message:

"`aes\_string()` was deprecated in ggplot2 3.0.0.

Please use tidy evaluation idioms with `aes()`.

See also `vignette("ggplot2-in-packages")` for more information.

The deprecated feature was likely used in the [UpSetR](#) package.

Please report the issue to the authors."

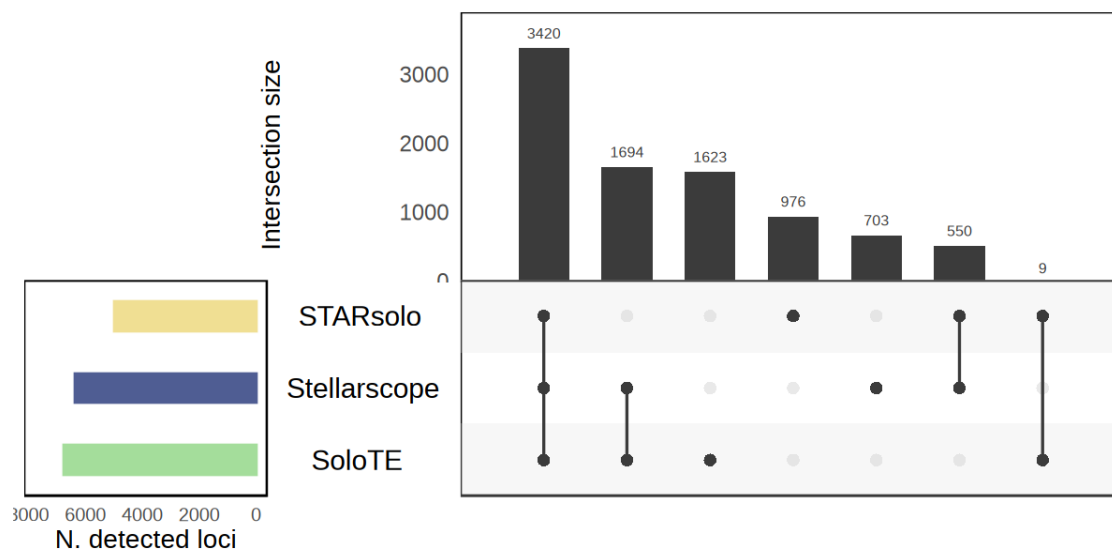
Warning message:

"Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.

Please use `linewidth` instead.

The deprecated feature was likely used in the [UpSetR](#) package.

Please report the issue to the authors."



```

[34]: library(eulerr)

# remove NAs
sum(is.na(detectedList[["SoloTE"]])) # ?
detectedList[["SoloTE"]][is.na(detectedList[["SoloTE"]])] <- (1:sum(is.
  ↪na(detectedList[["SoloTE"]]))
sum(is.na(detectedList[["SoloTE"]]))

# Convert lists to sets and compute intersections
fit <- euler(detectedList)

# Plot with proportional areas
plot(fit,
      fills = colorTools,
      alpha = 0.5,
      labels = list(font = 2, cex = 1.5),      # <-- increase set label size
      quantities = list(cex = 1.4),           # <-- increase count size
      main.cex = list(cex=2),
      edges = list(col = "white", lwd = 2),
      # labels = TRUE,
      # quantities = TRUE,
      main = "Overlap across detected TEs")

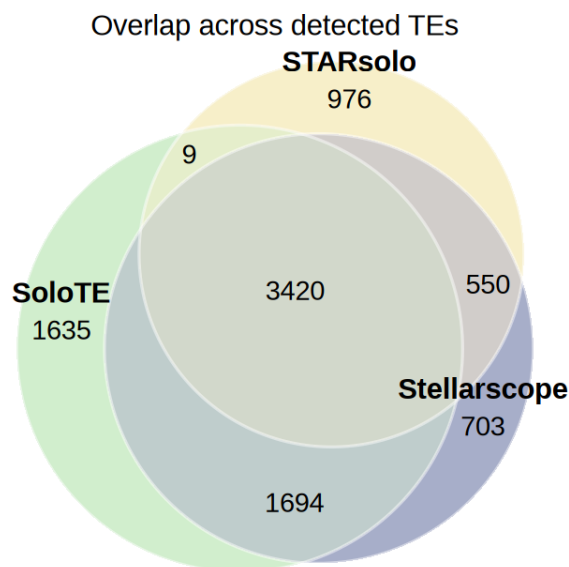
pdf(paste0("figures_", dataset_id, "/overlap_euler_withSTAR.pdf"), width=8,
  ↪height=8)
plot(fit,
      fills = colorTools,
      alpha = 0.5,
      labels = list(font = 2, cex = 1.5),      # <-- increase set label size
      quantities = list(cex = 1.4),           # <-- increase count size
      main.cex = list(cex=2),
      # labels = TRUE,
      # quantities = TRUE,
      main = "Overlap across detected TEs")
dev.off()

```

13

0

agg__record__1909893419: 2



```
[34]: # # Number of TEs detected above a certain threshold of cells ( 5% of the
      ↪ dataset)

# cellNum <- round(length(Cells(objTE_soloTE))*0.05)
# # save lists of TEs expressed in each sample
# nTEs_soloTE <- list()
# TEs_soloTE <- list()

# for(ident in unique(objTE_soloTE$orig.ident)){
#   mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],
      ↪ layer="counts")
#   nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
#   TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
# }
# TEs_soloTE_df <- stack(TEs_soloTE)
# colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

# nTEs_stellarscope <- list()
# TEs_stellarscope <- list()
# for(ident in unique(objTE_stellarscope$orig.ident)){
#   mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
      ↪ ident==ident], layer="counts")
#   nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
#   TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
# }
# TEs_stellarscope_df <- stack(TEs_stellarscope)
# colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")
```

```

# nTEs_star <- list()
# TEs_star <- list()
# for(ident in unique(objTE_STARsolo$orig.ident)){
#   mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident],
#     ↪layer="counts")
#   nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
#   TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
# }
# TEs_star_df <- stack(TEs_stellarscope)
# colnames(TEs_star_df) <- c("locus", "orig.ident")

# options(repr.plot.width=4, repr.plot.height=5)
# # N TEs per tool
# df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE),
#     ↪rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
#   cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope),
#     ↪rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
#   cbind(unlist(nTEs_star), names(nTEs_star),
#     ↪rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
# )

# colnames(df) <- c("n_TEs", "Sample", "Tool")
# df <- as.data.frame(df)
# df$n_TEs <- as.numeric(df$n_TEs)
# df$Sample <- factor(df$Sample, levels=unique(df$Sample))
# df
# ggplot(df, aes(x=Tool, fill=Tool, color=Tool, y=n_TEs)) +
#   geom_col(position = "dodge") +
#   scale_fill_manual(values=colorTools)+
#   scale_color_manual(values=colorTools)+
#   ggtitle(paste0("Loci detected in 5% of cells")) +
#   xlab("") +
#   ylab("N. detected TEs") +
#   theme_minimal() +
#   theme(text=element_text(size=18),
#     plot.title = element_text(size=18, hjust=0.5),
#     plot.subtitle = element_text(size=18, hjust=0.5),
#     axis.text.x = element_text(size=15, angle=45, hjust=1))
# ggsave(paste0("figures_", dataset_id, "/
#     ↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_thr5percCells.pdf"), device =
#     ↪"pdf", width = 5, height=5)

```

```

[35]: options(repr.plot.width=5, repr.plot.height=7)
TEs_soloTE_df$age <- conversionTable[match(TEs_soloTE_df$locus,
  ↪conversionTable$soloteID), "ageClass"]
TEs_stellarscope_df$age <- conversionTable[match(TEs_stellarscope_df$locus,
  ↪conversionTable$stellarscopeID), "ageClass"]

```

```

TEs_star_df$age <- conversionTable[match(TEs_star_df$locus,
↳conversionTable$stellarscopeID), "ageClass"]

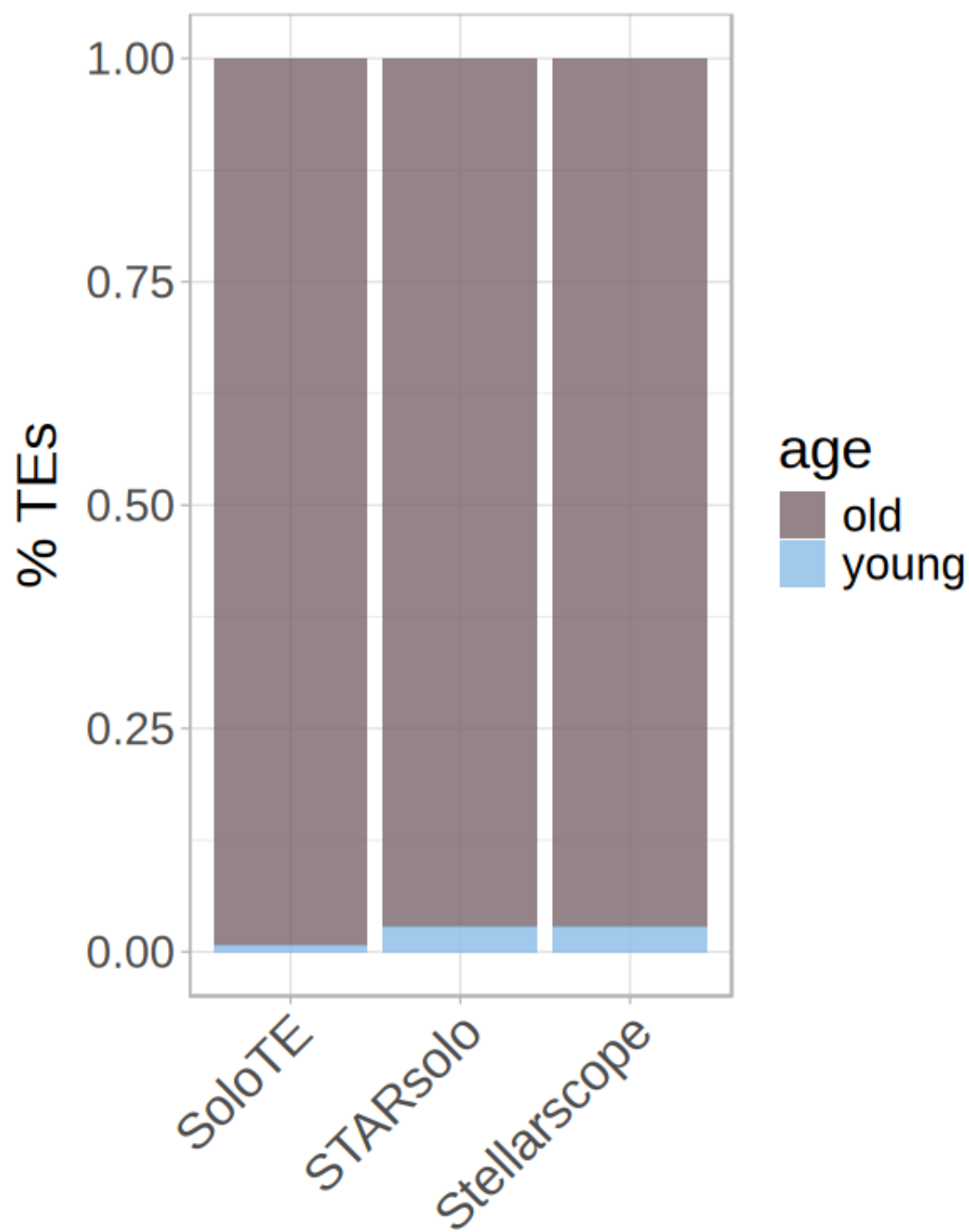
TEs_soloTE_df$tool <- "SoloTE"
TEs_stellarscope_df$tool <- "Stellarscope"
TEs_star_df$tool <- "STARsolo"
df <- rbind(TEs_soloTE_df, TEs_stellarscope_df, TEs_star_df)

df <- df[!is.na(df$age),] # why are there NAs? Didn't I remove TEs not in the
↳annotation beforehand?

ggplot(df, aes(x=tool, fill=age))+
  geom_bar(position="fill", alpha = 0.8) +
  scale_fill_manual(values=colorAge) +
  ggtitle("") +
  xlab("") +
  ylab("% TEs") +
  theme_light() +
  theme(text=element_text(size=20),
        strip.text.x = element_text(size=22, hjust=0.5),
        strip.background = element_rect(fill="#7E7E7E"),
        axis.text.x = element_text(size=18, angle=45, hjust=1))

ggsave(paste0("figures_", dataset_id, "/
↳nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_age_tool_celltype.pdf"), device=
↳ "pdf", width=5, height=7)

```



```

[36]: options(repr.plot.width=8, repr.plot.height=7)
soloTEageFamily <- table(conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "ageClass"],
  conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "class"])
soloTEageFamilyOld <-
  ↪cbind(soloTEageFamily["old",], colnames(soloTEageFamily), "old", "SoloTE")
colnames(soloTEageFamilyOld) <- c("nTEs", "superfamily", "age", "tool")
soloTEageFamilyYoung <-
  ↪cbind(soloTEageFamily["young",], colnames(soloTEageFamily), "young", "SoloTE")
colnames(soloTEageFamilyYoung) <- c("nTEs", "superfamily", "age", "tool")
soloTEageFamily <- rbind(soloTEageFamilyOld, soloTEageFamilyYoung)
soloTEageFamily <- as.data.frame(soloTEageFamily)
soloTEageFamily$nTEs <- as.numeric(soloTEageFamily$nTEs)

stellarscopeageFamily <-
  ↪table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"],
  conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "class"])
stellarscopeageFamilyOld <-
  ↪cbind(stellarscopeageFamily["old",], colnames(stellarscopeageFamily), "old", "Stellarscope")
colnames(stellarscopeageFamilyOld) <- c("nTEs", "superfamily", "age", "tool")
stellarscopeageFamilyYoung <-
  ↪cbind(stellarscopeageFamily["young",], colnames(stellarscopeageFamily), "young", "Stellarscope")
colnames(stellarscopeageFamilyYoung) <- c("nTEs", "superfamily", "age", "tool")
stellarscopeageFamily <- rbind(stellarscopeageFamilyOld,
  ↪stellarscopeageFamilyYoung)
stellarscopeageFamily <- as.data.frame(stellarscopeageFamily)
stellarscopeageFamily$nTEs <- as.numeric(stellarscopeageFamily$nTEs)

ageFamilyDf <- rbind(soloTEageFamily, stellarscopeageFamily)
ageFamilyDf$superfamily <- gsub(ageFamilyDf$superfamily, pattern="\\?",
  ↪replacement="")

ageFamilyDf <- ageFamilyDf[ageFamilyDf$nTEs>0,]

ggplot(ageFamilyDf, aes(x=tool, fill=superfamily, y=nTEs)) +
  facet_wrap(~age) +
  geom_col(position="fill") +
  ↪#scale_fill_manual(values=adjustcolor(c("#5E50A1", "#3787BC", "#67C1A3", "#AADCA2", "#E6F596"),
  ↪alpha.f = 0.8)) +
  scale_fill_brewer(palette = "Spectral", direction = -1)+

```



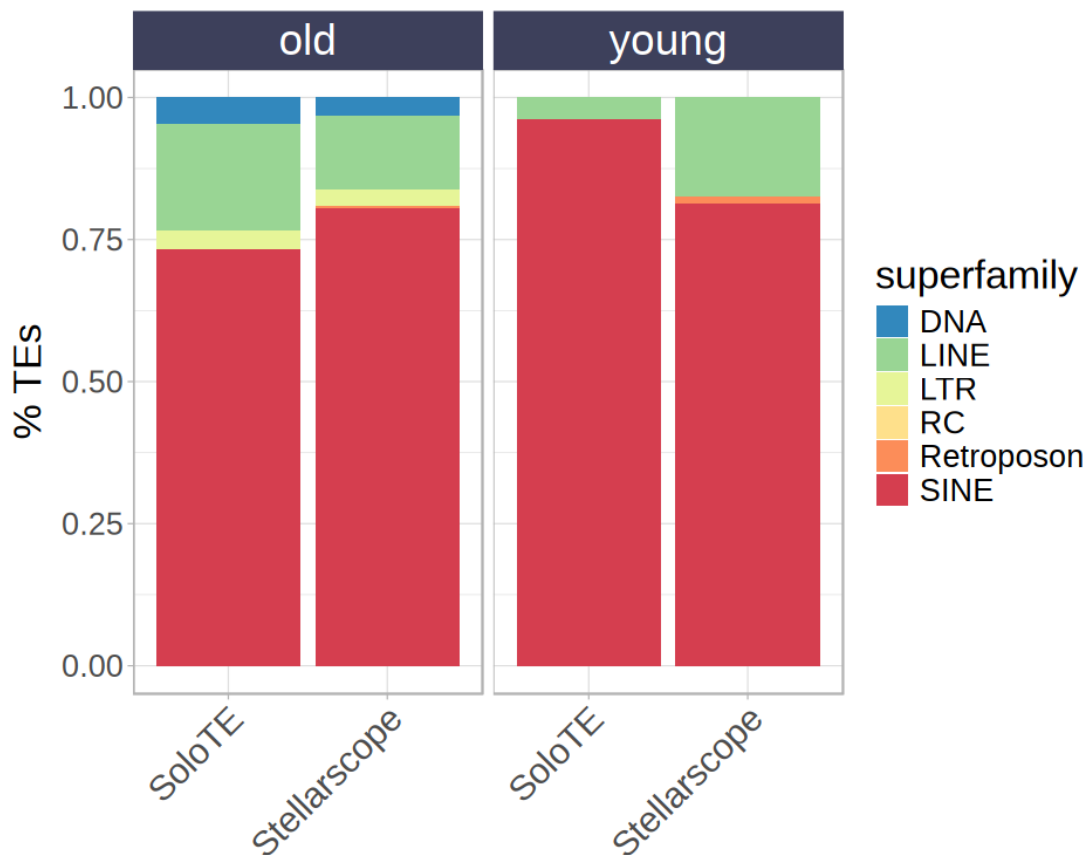
```

#palette = "Spectral", direction = -1)+
# scale_color_manual(values=c("#81B29A", "#9181A6"))+
ggtitle("") +
xlab("") +
ylab("% TEs") +
theme_light() +
theme(text=element_text(size=20),
      strip.text.x = element_text(size=22, hjust=0.5),
      strip.background = element_rect(fill="#3d405b", #"#e78063"
      axis.text.x = element_text(size=18, angle=45, hjust=1))

ggsave(paste0("figures_", dataset_id, "/
↳ nTEs_SoloTEvsStellarscope_barplot_age_order.pdf"), device = "pdf")

```

Saving 6.67 x 6.67 in image



```

[37]: options(repr.plot.width=4, repr.plot.height=6)

nOldSoloTE <- table(conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "ageClass"])[ "old"]
nYoungSoloTE <- table(conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "ageClass"])[ "young"]
nOldStellarscope <- table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "old"]
nYoungStellarscope <- table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "young"]
nOldSTARsolo <- table(conversionTable[match(Features(objTE_STARsolo),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "old"]
nYoungSTARsolo <- table(conversionTable[match(Features(objTE_STARsolo),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "young"]

table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"])

df <- rbind( c(nOldSoloTE , nOldSoloTE / (nOldSoloTE + nYoungSoloTE), "old",
  ↪"SoloTE"),
             c(nYoungSoloTE, nYoungSoloTE / (nOldSoloTE + nYoungSoloTE),
  ↪"young", "SoloTE"),
             c(nOldStellarscope, nOldStellarscope / (nOldStellarscope +
  ↪nYoungStellarscope), "old", "Stellarscope"),
             c(nYoungStellarscope, nYoungStellarscope / (nOldStellarscope +
  ↪nYoungStellarscope), "young", "Stellarscope"),
             c(nOldSTARsolo, nOldSTARsolo / (nOldSTARsolo + nYoungSTARsolo),
  ↪"old", "STARsolo"),
             c(nYoungSTARsolo, nYoungSTARsolo / (nOldSTARsolo +
  ↪nYoungSTARsolo), "young", "STARsolo")
           )
df <- as.data.frame(df)

colnames(df) <- c("n_TEs", "n_TEs_prop", "AgeClass", "Tool")
df$n_TEs <- as.numeric(df$n_TEs)
df$n_TEs_prop <- as.numeric(df$n_TEs_prop)
df$n_TEs_perc <- as.numeric(df$n_TEs_prop) * 100

ggplot(df[df$AgeClass=="young",], aes(x=Tool, fill=Tool, color=Tool,
  ↪y=n_TEs_perc)) +
  geom_col(position = "stack") +
  scale_fill_manual(values=colorTools)+

```

```

scale_color_manual(values=colorTools)+
ggtitle("") +
xlab("") +
ylab("% young TEs") +
theme_minimal() +
theme(text=element_text(size=18),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.x = element_text(size=18, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/barplot_percYoungTEs.pdf"), device = "pdf",
      width = 4, height=6)

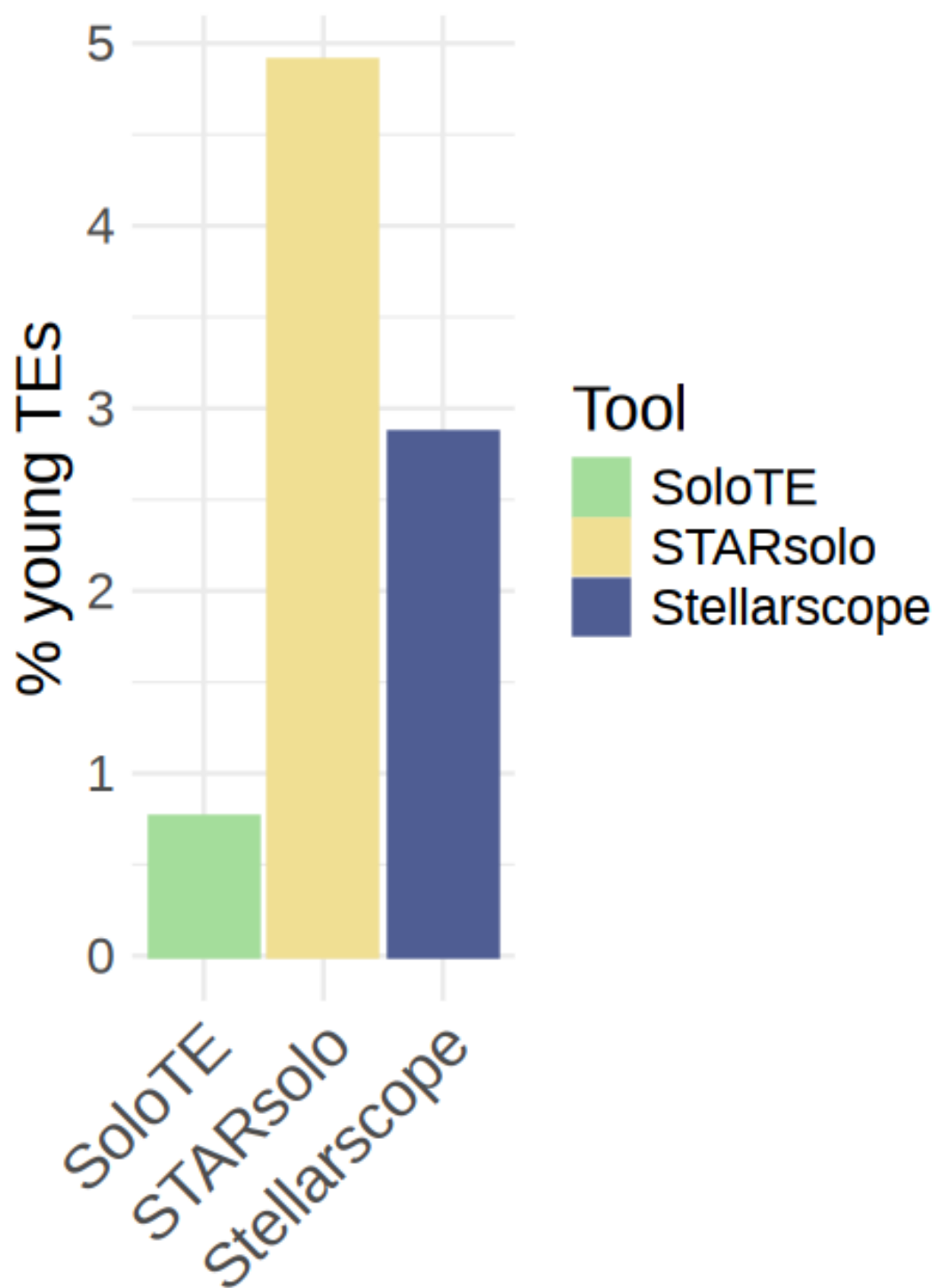
ggplot(df[df$AgeClass=="young",], aes(x=Tool, fill=Tool, color=Tool, y=n_TEs))
+
geom_col(position = "stack") +
scale_fill_manual(values=colorTools)+
scale_color_manual(values=colorTools)+
ggtitle("") +
xlab("") +
ylab("N. young TEs") +
theme_minimal() +
theme(text=element_text(size=18),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.x = element_text(size=18, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/barplot_nYoungTEs.pdf"), device = "pdf",
      width = 4, height=6)

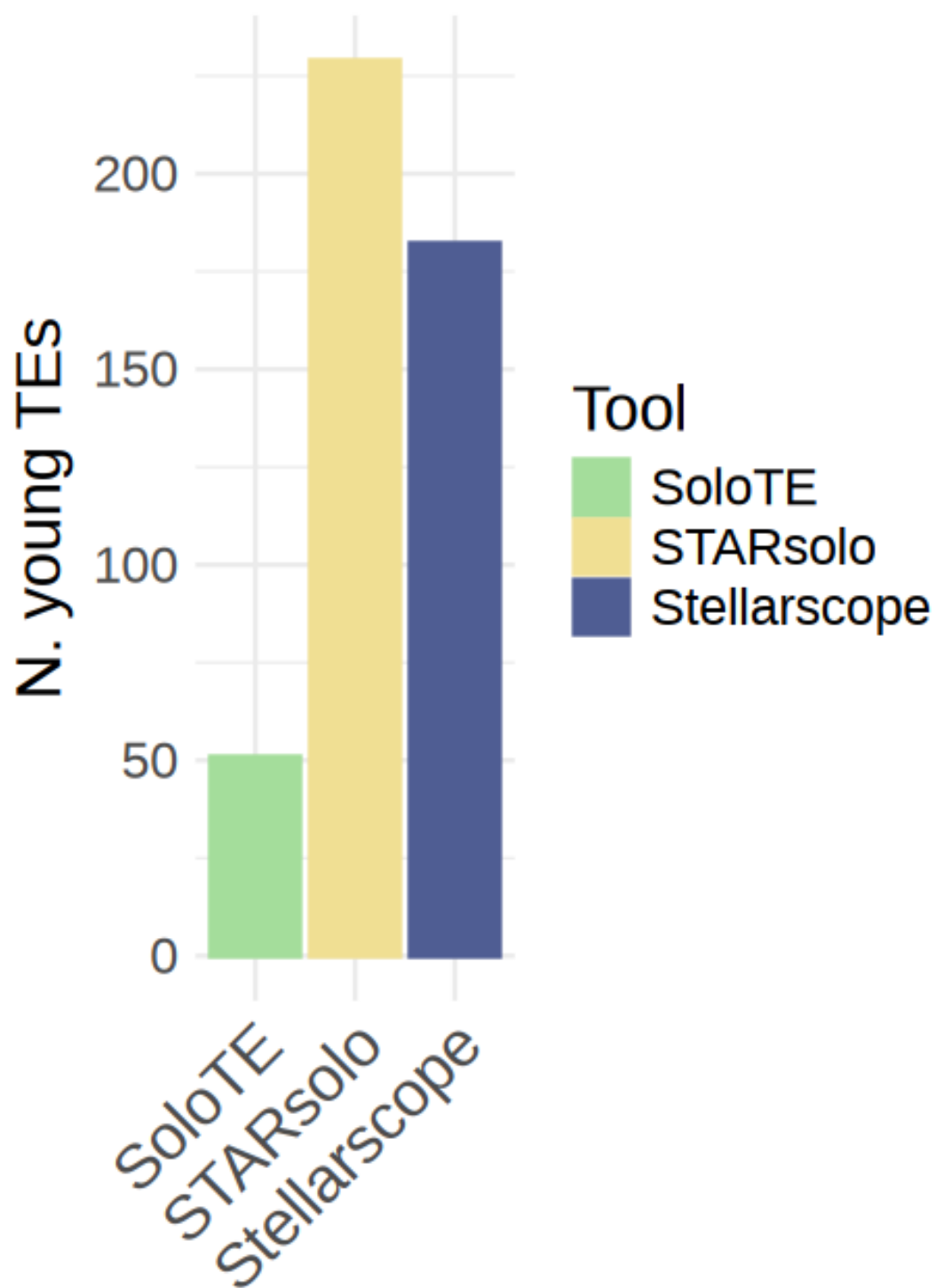
```

```

old young
6154 182

```





6 Celltype purity

```
[5]: objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:13, k.param = 20, return.neighbor = T)

purity <- sapply(Cells(objTE_stellarscope), FUN = function(cell){
  n_cells <- TopNeighbors(objTE_stellarscope@neighbors$RNA.nn, cell, n = 20)
  celltypes <- objTE_stellarscope@meta.data[n_cells, 'celltype']
  label <- objTE_stellarscope@meta.data[cell, 'celltype']
  perc_label <- length(celltypes[celltypes==label]) / 20
})
df <- as.data.frame(cbind(purity, "Stellarscope"))
colnames(df) <- c("purity", "tool")
df$celltype <- objTE_stellarscope@meta.data[rownames(df), 'celltype']
df$purity <- as.numeric(df$purity)

objTE_soloTE <- FindNeighbors(objTE_soloTE, dims = 1:13, k.param = 20, return.
  ↪neighbor = T)
objTE_soloTE$celltype <- celltypes$V2[match(Cells(objTE_soloTE), celltypes$V1)]

purity <- sapply(Cells(objTE_soloTE), FUN = function(cell){
  label <- objTE_soloTE@meta.data[cell, 'celltype']
  n_cells <- TopNeighbors(objTE_soloTE@neighbors$RNA.nn, cell, n = 20)
  celltypes <- objTE_soloTE@meta.data[n_cells, 'celltype']
  label <- objTE_soloTE@meta.data[cell, 'celltype']
  perc_label <- length(celltypes[celltypes==label]) / 20
})
df2 <- as.data.frame(cbind(purity, "SoloTE"))
colnames(df2) <- c("purity", "tool")
df2$celltype <- objTE_soloTE@meta.data[rownames(df2), 'celltype']
df2$purity <- as.numeric(df2$purity)

objTE_STARsolo <- FindNeighbors(objTE_STARsolo, dims = 1:13, k.param = 20,
  ↪return.neighbor = T)

purity <- sapply(Cells(objTE_STARsolo), FUN = function(cell){
  label <- objTE_STARsolo@meta.data[cell, 'celltype']
  n_cells <- TopNeighbors(objTE_STARsolo@neighbors$RNA.nn, cell, n = 20)
  celltypes <- objTE_STARsolo@meta.data[n_cells, 'celltype']
  label <- objTE_STARsolo@meta.data[cell, 'celltype']
  perc_label <- length(celltypes[celltypes==label]) / 20
```

```

})
df3 <- as.data.frame(cbind(purity, "STARsolo"))
colnames(df3) <- c("purity", "tool")
df3$celltype <- objTE_STARsolo@meta.data[rownames(df2), 'celltype']
df3$purity <- as.numeric(df3$purity)

objTE_stellarscope_celltype <- FindNeighbors(objTE_stellarscope_celltype, dims = 1:13, k.param = 20, return.neighbor = T)

purity <- sapply(Cells(objTE_stellarscope_celltype), FUN = function(cell){
  n_cells <- TopNeighbors(objTE_stellarscope_celltype@neighbors$RNA.nn, cell, n_neighbors = 20)
  celltypes <- objTE_stellarscope_celltype@meta.data[n_cells, 'celltype']
  label <- objTE_stellarscope_celltype@meta.data[cell, 'celltype']
  perc_label <- length(celltypes[celltypes==label]) / 20
})
df4 <- as.data.frame(cbind(purity, "Stellarscope_byCelltype"))
colnames(df4) <- c("purity", "tool")
df4$celltype <- objTE_stellarscope@meta.data[rownames(df4), 'celltype']
df4$purity <- as.numeric(df4$purity)

df_purity <- rbind(df, df2, df3, df4)

```

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

```

[6]: df_purity$celltype <- gsub(pattern="CD14pos_Monocytes", replacement = "Monocytes_CD14pos", df_purity$celltype)
df_purity$celltype <- gsub(pattern="FCGR3Apos_Monocytes", replacement = "Monocytes_FCGR3Apos", df_purity$celltype)

```



```

df_purity$celltype <- gsub(pattern="Naive_CD4_T", replacement = "T_CD4_Naive",
  ↪df_purity$celltype)
df_purity$celltype <- gsub(pattern="Memory_CD4_T", replacement =
  ↪"T_CD4_Memory", df_purity$celltype)
df_purity$celltype <- gsub(pattern="CD8A", replacement = "T_CD8",
  ↪df_purity$celltype)
df_purity$celltype <- gsub(pattern="Megakaryocytes", replacement = "Platelet",
  ↪df_purity$celltype)

PBMCelltypeColors <- c("B"="#91bcca",
  "Monocytes_CD14pos"="#3d405b",
  "T_CD8"="#f2cc8f",
  "Dendritic"="#c492a7",
  "Monocytes_FCGR3Apos"="#e78063",
  "#Platelet"="#c4b3ab",
  "T_CD4_Memory"="#81b29a",
  "T_CD4_Naive"="#4d7362",
  "NK"= "#88535a")

```

```

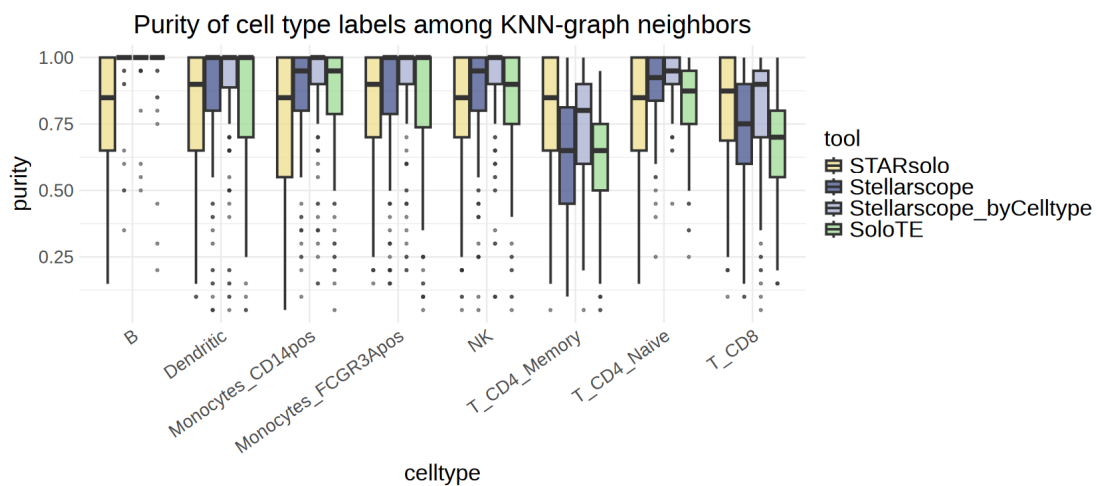
[7]: colorTools = c("SoloTE"="#A4DD9B", #6fc69d",
  "Stellarscope"="#4f5d93",
  "Stellarscope_byCelltype"="#ACB4D3",
  "STARsolo"="#f0df93")

df_purity$tool <- factor(df_purity$tool, levels =
  ↪c("STARsolo", "Stellarscope", "Stellarscope_byCelltype", "SoloTE"))

options(repr.plot.width=14, repr.plot.height=6)

ggplot(df_purity) + aes(y=purity, x=celltype, fill=tool) +
  ↪geom_boxplot(linewidth = 0.8, alpha=0.8, outlier.size=0.7, outlier.alpha = 0.
  ↪5) +
  ggtitle("Purity of cell type labels among KNN-graph neighbors") +
  scale_fill_manual(values=colorTools)+
  theme_minimal() +
  theme(text=element_text(size=20),
    plot.title = element_text(size=23, hjust=0.5),
    legend.text = element_text(size=20),
    axis.text.x = element_text(size=16, angle=35, hjust=1))
ggsave(paste0("figures_", dataset_id, "/
  ↪celltype_purity_knn_soloTEvsStellarscopevsSTAR_noPlatelet.pdf"), width=14,
  ↪height=6 )

```



[]:

[]: